COMPUTER VISION

LECTURE NOTES

IMAGE FORMATION
GEOMETRY
RECOGNITION
PERCEPTION
APPLIED VISION



NOTES PREPARED BY

Matt Clifford Trinidad Ramos

Contents

Lecture 1: Introduction: Scope, Challenges, and Applications 1

1 Computer Vision: Purpose and Scope 2
- 1.1 What Computer Vision Tries to Recover . . . 2
- 1.2 Why Visual Data Matter . . . 3
- 1.3 Vision and Graphics . . . 3

2 Why Vision Is Difficult 4
- 2.1 An Ill-Posed Inverse Problem . . . 4
- 2.2 Nuisance Parameters . . . 4
- 2.3 Context and Intra-Class Variation . . . 5
- 2.4 Visual Cues . . . 5
- 2.5 Scale and Complexity . . . 6

3 Historical Development and Changing Emphasis 6

4 Measurement and Semantic Understanding 8
- 4.1 Vision as a Measurement Device . . . 8
- 4.2 Vision as a Source of Semantic Information . . . 8

5 Core Areas of Computer Vision 9
- 5.1 Early Vision: Image Formation and Processing . . . 9
- 5.2 Mid-Level Vision: Grouping, Fitting, and Tracking . . . 10
- 5.3 Multi-View Geometry . . . 10
- 5.4 Recognition . . . 11
- 5.5 High-Level Geometric Vision . . . 11
- 5.6 Advanced Topics . . . 11
- 5.7 Image Collections and Image-Based Rendering . . . 11

6 Applications of Computer Vision 12

7 Computer Vision and Machine Learning 15
- 7.1 The Role of Machine Learning . . . 15
- 7.2 Related Disciplines . . . 16
- 7.3 Active Research Topics . . . 16
- 7.4 Research Publications . . . 16
- 7.5 Further Reading . . . 16

Lecture 1 Quick Review 17

Lecture 2: Image Acquisition and Digitalization 18

8 Image Formation and Camera Acquisition 19
8.1 Illumination and Surface Properties . . . 19
8.2 A Spectral Image-Formation Model . . . 19
8.3 Perspective and Orthographic Projection . . . 20
8.4 Color Sensors . . . 21

9 Digital Images: Sampling, Quantization, and Resolution 21
9.1 From a Continuous Image to a Digital Image . . . 21
9.2 Quantization and Bit Depth . . . 22
9.3 Digital Image Types . . . 23
9.4 Pixels, Memory, and Spatial Resolution . . . 23

10 Pixel Relationships and Image Resampling 25
10.1 Neighborhoods . . . 25
10.2 Connectivity, Adjacency, and Paths . . . 25
10.3 Distance Measures . . . 26
10.4 Interpolation . . . 26

Lecture 2 Quick Review 29

Lecture 3: Color, Perception, and Texture 30

11 Color: Light and Human Vision 31
11.1 The Nature of Color . . . 31
11.2 The Eye as a Sensing System . . . 31
11.3 Spectral Filtering and Metamers . . . 33

12 Color Matching and Color Spaces 34
12.1 Trichromacy and Color Matching . . . 34
12.2 Grassmann's Laws . . . 34
12.3 Color-Matching Functions as a Linear Transform . . . 35
12.4 RGB and CIE XYZ . . . 35
12.5 Toward Perceptual Uniformity . . . 35
12.6 HSV . . . 36

13 Color Constancy and White Balance 37
13.1 Constancy and Context . . . 37
13.2 Chromatic Adaptation . . . 38
13.3 White Balance . . . 39

14 Color in Computer Vision and Texture 40
14.1 Uses of Color . . . 40
14.2 Structural and Statistical Texture . . . 40
14.3 Local Binary Patterns . . . 41
14.4 Image Data Structure . . . 41

Lecture 3 Quick Review 42

Lecture 4: Histograms and Intensity Transformations 43

15 Histograms and Point Transformations 44
15.1 Intensity Histograms . . . 44
15.2 Histograms of Color Images . . . 45
15.3 Point Transformations . . . 45

16 Histogram Equalization, Projection, and Plateau Methods 46
16.1 Histogram Equalization . . . 46
16.2 Histogram Projection . . . 46
16.3 Plateau Equalization . . . 47

17 Histogram Specification and Contrast Stretching 48
17.1 Histogram Specification . . . 48
17.2 Contrast Stretching . . . 48

18 Applications of Histogram Processing 49
18.1 Thresholding . . . 49
18.2 Normalization and Histogram Matching . . . 49
18.3 Medical-Image Examples . . . 49
18.4 High-Dynamic-Range Display . . . 50

Lecture 4 Quick Review 51

Lecture 5: Spatial Filters and Image Enhancement 52

19 Spatial Filtering Foundations 53
19.1 Neighborhood Operations . . . 53
19.2 The Spatial Filtering Process . . . 53
19.3 Correlation and Convolution . . . 54
19.4 Image Borders . . . 54

20 Smoothing Spatial Filters 54
20.1 Averaging Filters . . . 54
20.2 Weighted Smoothing . . . 55
20.3 Smoothing as Preprocessing . . . 55
20.4 Averaging and Median Filtering . . . 55

21 Sharpening and Spatial Derivatives 56
21.1 First and Second Discrete Derivatives . . . 56
21.2 The Laplacian . . . 56
21.3 Unsharp Masking and Highboost Filtering . . . 57
21.4 First-Derivative Filtering and the Gradient . . . 58
21.5 Sobel Operators . . . 58

22 Combining Spatial Enhancement Methods 59
22.1 Bone-Scan Enhancement Pipeline . . . 59

Lecture 5 Quick Review 61

LECTURE 1

Introduction: Scope, Challenges, and Applications

From visual measurements to descriptions of the world.

Scope · Visual cues · Geometry · Recognition

Computer Vision — Marco Raoul Marini

Notes prepared by Matt Clifford Trinidad Ramos

CHAPTER 1

Computer Vision: Purpose and Scope

Computer vision aims to make computational systems extract properties of the three-dimensional world from one or more images. The desired result may be a physical measurement, an object or scene category, a location, a spatial relation, or another description that is useful for a task.



Figure 1.1. A visual metaphor for the central goal of computer vision: enabling computational systems to interpret visual input.



Figure 1.2. The source's computer-vision pipeline: visual input becomes an interpretation of the world through sensing and interpretation.

1.1 What Computer Vision Tries to Recover

> **Computer Vision**
>
> Given one image or several images, computer vision extracts properties of the three-dimensional world represented by those images.

A traffic scene illustrates the range of possible outputs. From the same image, a system may be asked to determine the number and type of vehicles, the location of the closest obstacle, the level of congestion, or

the location and nature of the observed scene.

> **Scene understanding from a single photograph**
>
> A visual scene may support several simultaneous questions: where the image was captured, how many people are present, what they are doing, what objects are supporting or surrounding them, and whether the situation contains an unusual or humorous interaction. These questions require more than measuring isolated pixels.

1.2 Why Visual Data Matter

Images and video occur in personal photo collections, movies, news, sport, surveillance and security, medical imaging, scientific imaging, and many other settings. Their abundance makes visual analysis useful in both everyday systems and specialized scientific applications.

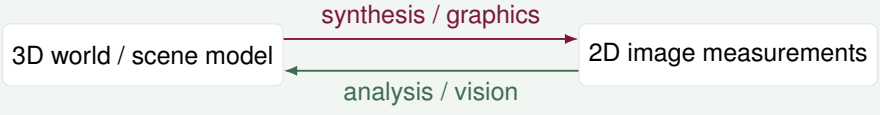
The broad motivation for computer vision includes safety, health technology, security, comfort, entertainment, and accessibility. The same field therefore spans both measurement-oriented problems and semantic interpretation.



Figure 1.3. Representative domains in which images and video occur: personal media, movies and news, surveillance, and scientific or medical imaging.

1.3 Vision and Graphics

Computer graphics and computer vision address opposite directions of the same relationship. Graphics synthesizes images from a model of the world; vision analyzes images to infer properties of the world.

> **Vision and graphics: opposite directions**
>
> 
>

The central computer-vision problem is inverse: a three-dimensional world produces a two-dimensional image, but the system must reason from the image back toward properties of that world.

Chapter summary. Computer vision converts visual measurements into information about the world. It supports both geometric measurement and semantic interpretation, and it is closely related to computer graphics through the analysis–synthesis relationship.

CHAPTER 2

Why Vision Is Difficult

Vision is difficult because an image is an incomplete and ambiguous measurement. The world is three-dimensional, images are two-dimensional, and the appearance of the same object can vary dramatically with conditions.

2.1 An Ill-Posed Inverse Problem

The real world is much more complex than what can be measured in an image. Projection from 3D to 2D removes information, so image formation cannot simply be inverted. A vision system must infer hidden properties from evidence that is incomplete and often ambiguous.

2.2 Nuisance Parameters

Several factors change image appearance without necessarily changing the identity of the object or scene.

Table 2.1. Major sources of variation in visual appearance.

Factor	Effect on the observed image
Illumination	The same surface can appear different when lighting changes.
Object pose	Articulated or movable objects can change configuration.
Clutter	Irrelevant surrounding structure can interfere with interpretation.
Occlusion	Part of an object can be hidden by another object.
Intra-class appearance	Different members of the same category can look different.
Viewpoint	Appearance changes as the camera observes the object from a different direction.
Scale	The same object occupies different numbers of pixels at different distances or resolutions.
Deformation	Non-rigid objects may change shape.
Background clutter	Object evidence must be separated from a complicated background.

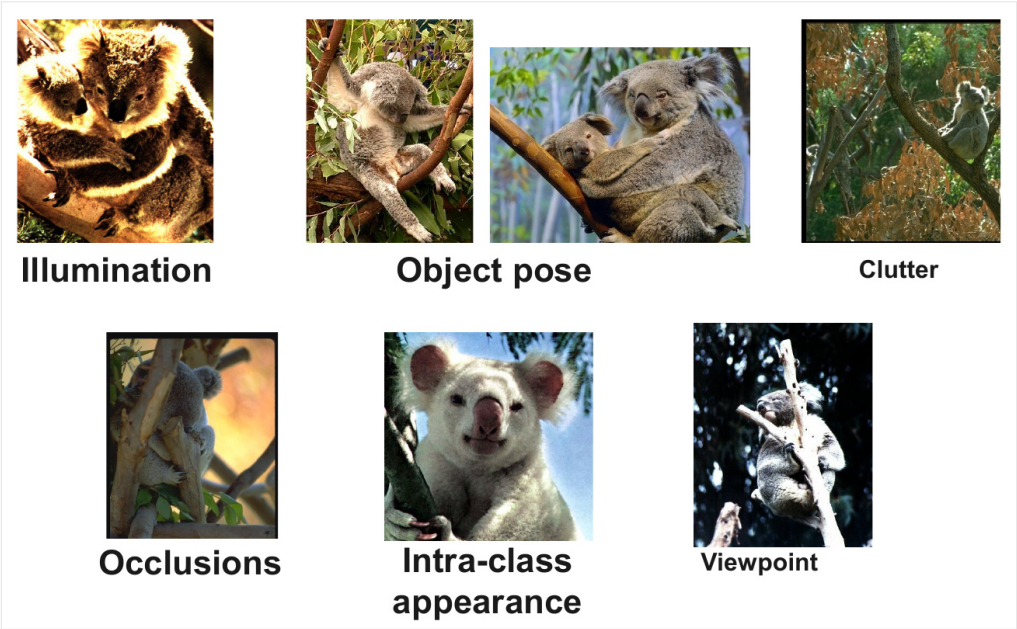


Figure 2.1. The same object category can change appearance because of illumination, pose, clutter, occlusion, intra-class differences, and viewpoint.

2.3 Context and Intra-Class Variation

Intra-class variation means that objects belonging to the same category may have very different visual appearances. Context is also important: a local region that is ambiguous in isolation can become easier to interpret when its surroundings are considered.

Viewpoint, illumination, scale, deformation, occlusion, and background clutter are therefore not exceptional cases. They are normal conditions that a useful vision system must tolerate.

2.4 Visual Cues

Images are difficult to interpret, but they also reveal structure through visual cues. The task is to exploit those cues rather than treating all variation as noise.

2.4.1 Depth and Ordering Cues

Linear perspective and aerial perspective provide information about depth. Occlusion provides depth ordering: when one object hides part of another, the occluding object is interpreted as being in front.

2.4.2 Shape and Lighting Cues

Texture gradients can support shape interpretation. Shading and cast shadows also provide information about shape, position, and illumination.

2.4.3 Grouping Cues

Similarity in color, texture, or proximity can group image elements into coherent structures. Common fate provides another grouping cue: image elements moving together are likely to belong together.

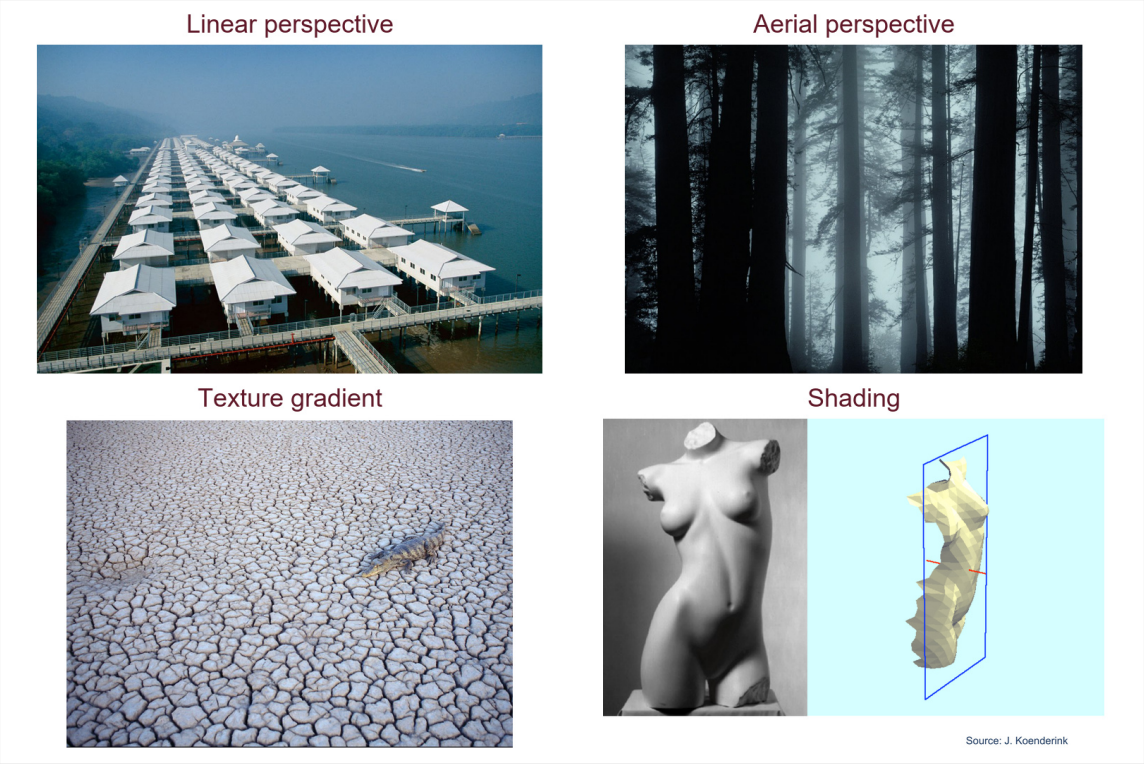


Figure 2.2. Examples of visual cues: linear perspective, aerial perspective, texture gradients, and shading.

2.5 Scale and Complexity

The computational scale of vision is large. A single image may contain thousands to millions of pixels. The space of human-recognizable object categories was described as roughly 3,000–30,000 categories, and articulated objects such as humans can involve more than 30 degrees of freedom. Visual data on the web add images, animated images, and video at very large scale.

A biological comparison reinforces the difficulty: approximately half of the cerebral cortex in primates has been associated with visual processing in the cited neurophysiological literature.

> **Chapter summary.** Vision is hard because the mapping from the world to an image is many-to-one and because appearance changes with nuisance parameters. Depth, shape, lighting, and grouping cues provide information that helps recover structure despite this ambiguity.

CHAPTER 3

Historical Development and Changing Emphasis

The history of computer vision shows a recurring shift between geometry, statistical modeling, data, and learning. Early optimism was followed by increasingly sophisticated representations and a recognition that visual understanding is not a short-term engineering problem.

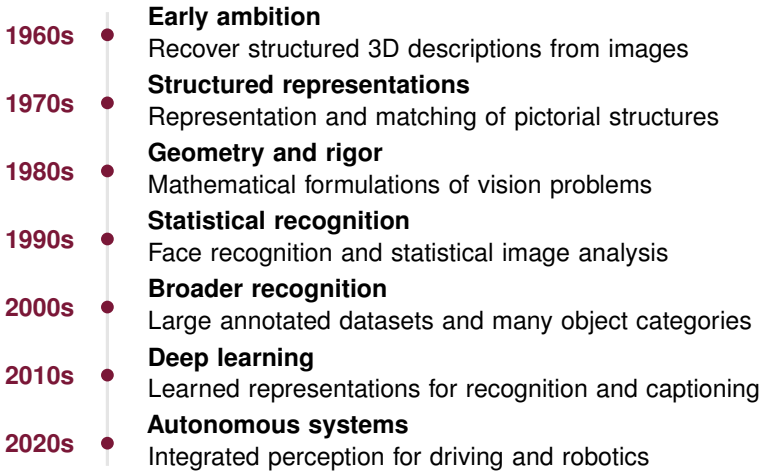


Figure 3.1. Source milestones reconstructed as a readable timeline; the following sections retain the historical examples.

3.1 The Early Ambition

In 1966, Marvin Minsky assigned a summer project whose goal was essentially to connect a camera to a computer and make the machine describe what it saw. Decades later, general visual understanding remained an open research problem.

Larry Roberts' 1963 MIT PhD work, *Machine Perception of Three-Dimensional Solids*, became an early landmark in interpreting synthetic scenes and recovering three-dimensional structure.

3.2 1970s: Structured Representations

Work in the 1970s made progress on selected images and structured visual representations. Fischler and Elschlager's 1973 work on the representation and matching of pictorial structures is a representative example.

3.3 1980s: Geometry and Mathematical Rigor

The 1980s saw artificial neural networks appear and then recede in influence while the field moved strongly toward geometry and more mathematically rigorous formulations.

3.4 1990s: Statistical Recognition

Face recognition became a prominent task, and statistical analysis became increasingly important in visual recognition.

3.5 2000s: Broader Recognition and Data

Recognition expanded to a broader set of categories. Large annotated datasets became increasingly important, and video processing grew as a major area of work.

3.6 2010s: Deep Learning

Deep learning returned to prominence. Representative milestones listed in the historical timeline include AlexNet (NIPS 2012), DeepFace (CVPR 2014), DeepPose (CVPR 2014), and *Show, Attend and Tell* (ICML 2015).

3.7 2020s: Autonomous Systems

Autonomous vehicles represent a major modern application direction in which recognition, geometry, motion, and real-time decision making must operate together.

> **Chapter summary.** Computer vision evolved from early synthetic-scene interpretation through structured representations, geometric rigor, statistical recognition, large annotated datasets, deep learning, and autonomous systems.

CHAPTER 4

Measurement and Semantic Understanding

Computer vision is useful both as a measurement technology and as a way to obtain semantic information. These two goals are complementary: one emphasizes geometry and physical structure, while the other emphasizes categories and meanings that matter to people.

4.1 Vision as a Measurement Device

Measurement-oriented vision includes real-time stereo, structure from motion, multi-view stereo, and three-dimensional reconstruction from collections of images. Space exploration provides a concrete example: visual systems can support terrain reconstruction, obstacle detection, and navigation.

> **Why learning still matters**
>
> For some physical measurements, other sensors can be more direct or accurate than cameras. However, visual navigation and semantic tasks still require learned or data-driven interpretation, because the goals of vision include both *what* is present and *where* it is.

4.2 Vision as a Source of Semantic Information

A street scene can support object labels, scene context, and qualitative spatial information.



Figure 4.1. Semantic interpretation can assign object and scene labels to visually meaningful regions.

Table 4.1. Examples of semantic information obtainable from one scene.

Level	Examples
Object categorization	sky, building, flag, face, banner, bus, wall, street lamp, cars
Scene / context	outdoor, city, traffic
Qualitative spatial information	horizontal, vertical, slanted, rigid moving object, non-rigid moving object

Semantic vision therefore includes scene type, spatial organization, and motion-related interpretation, not only isolated object labels. **Chapter summary.** Vision combines geometry and semantics: systems may need both *what* is present and *where* it is.

CHAPTER 5

Core Areas of Computer Vision

A useful organization of the field separates early vision, mid-level vision, multi-view geometry, recognition, and high-level or advanced topics. The categories are not completely independent, but they provide a practical map of the techniques studied in computer vision.

5.1 Early Vision: Image Formation and Processing

Early vision begins with image formation and basic image processing. Its topics include cameras, light, color, filtering, and edges.

5.1.1 The Physics of Imaging

Camera modeling addresses what a camera does and how camera position can be inferred. Light is studied in terms of measurement, interaction with surfaces, and the way image brightness is produced. Color is studied through its underlying mechanisms and through methods for describing and measuring it.

The broader topic list includes imaging geometry, pinhole cameras, lenses, camera parameters, and calibration.

5.1.2 Representing Small Image Patches

Small image neighborhoods are represented for three main reasons:

- to establish correspondence between points in different images;
- to detect sharp changes, commonly called edges;
- to represent texture through statistics of local image patterns.

Filter outputs can be interpreted as a dot-product-like comparison between a pattern and the image while the pattern is shifted across image locations. A strong response indicates that the local image resembles the pattern. Derivative-like filters are useful for detecting intensity changes.

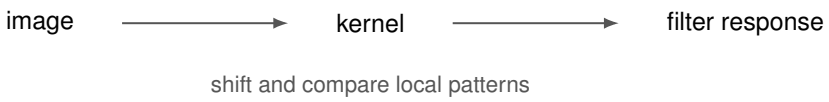


Figure 5.1. Conceptual filtering pipeline: a kernel is shifted across the image to produce local responses.

Image filtering and enhancement also include linear filters, convolution, smoothing, edge detection, and image pyramids.

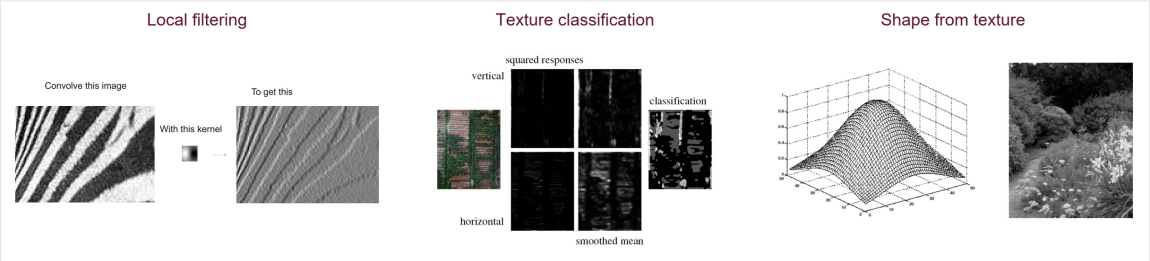


Figure 5.2. Early-vision examples: local filtering, texture classification from filter responses, and shape information derived from texture.

5.1.3 Texture

Many objects are distinguished by texture. Tigers, cheetahs, grass, and trees provide intuitive examples. Texture can be represented through statistics of filter outputs: coarse bar filters respond strongly to tiger-like stripes, spot filters to cheetah-like patterns, long narrow filters to grass-like structure, and extended spot-like responses to leaves.

Objects with different textures can be segmented, and variation in texture can provide a cue to three-dimensional shape.

5.2 Mid-Level Vision: Grouping, Fitting, and Tracking

Mid-level vision seeks coherent structure by grouping or fitting lower-level measurements. Fitting can include alignment and least-squares methods.

5.2.1 Segmentation

Segmentation breaks an image or video into useful pieces. Examples include finding video sequences corresponding to one shot and finding image components that are coherent in internal appearance.

5.2.2 Tracking

Tracking keeps track of a moving object through a long sequence of views.

5.3 Multi-View Geometry

Multi-view geometry studies relationships across several views. The core examples are stereo and projective structure from motion.

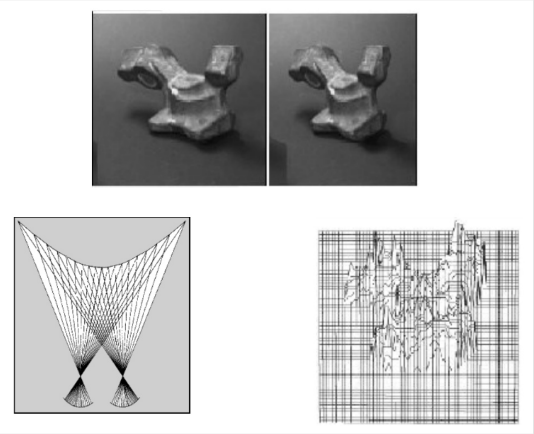


Figure 5.3. Stereo vision uses multiple views to recover geometric information that is not directly available from a single image.

5.4 Recognition

A representative recognition pipeline consists of patch description and matching, clustering and visual vocabularies, bag-of-features models, and classification.



Figure 5.4. Recognition pipeline from local patch description to classification.

5.5 High-Level Geometric Vision

High-level geometric vision studies relations between object geometry and image geometry. Its topics include model-based vision for finding the position and orientation of known objects, smooth surfaces and outlines, aspect graphs that describe how outlines change with viewpoint, and range data.

5.6 Advanced Topics

Representative advanced topics include segmentation, face detection, and articulated models. The broader topic list also includes image restoration, shape analysis, motion and optical flow, semantic detection, and image mosaics.

5.7 Image Collections and Image-Based Rendering

Two recurring application directions are finding images in large collections and image-based rendering. Image collections support searching and browsing. Image-based rendering addresses the difficulty of building geometric models that reproduce every real-world detail, such as surface weathering, by producing new pictures from existing images.

> **Chapter summary.** Early vision studies image formation and local measurements. Mid-level vision groups and fits those measurements. Multi-view geometry reasons across viewpoints, recognition assigns categories, and high-level topics connect image evidence to object geometry and complex applications.

CHAPTER 6

Applications of Computer Vision

Computer vision appears in systems ranging from document analysis to robotics and medical imaging. The examples below illustrate how the same core ideas recur in different practical settings.

6.1 Detection and Recognition

A basic detection task asks whether an image contains a person. The positive and negative examples used for this problem emphasize the need to distinguish target instances from images without the target.



Figure 6.1. Positive and negative examples illustrate the basic structure of a visual detection problem.

Face detection in digital cameras is a successful example: detected faces can be used to control autofocus, exposure, and flash output.

Smile detection and face recognition are further examples in which visual information is used to infer a specific facial state or identity.

6.2 Optical Character Recognition

Optical character recognition converts scanned documents into text. Related examples include handwritten digit recognition and automatic license-plate reading.

6.3 Retail and Mobile Recognition

Retail vision can detect and recognize products. A historical example is LaneHawk, in which a camera mounted at a checkout lane detects items under a shopping basket so that the cashier can verify the detected quantity.

Mobile-phone recognition extends the same idea to camera-equipped phones. Examples included Microsoft Research systems and Nokia Point & Find.

6.4 Biometrics and Password-Free Login

Vision-based biometrics includes face, iris, and fingerprint recognition. The identification of the "Afghan Girl" from iris patterns is an illustrative example of iris recognition. Fingerprint scanners and face-recognition systems also support login without a conventional password.

6.5 Special Effects and Interaction

Shape capture and motion capture are used in visual effects. Examples include work associated with *The Matrix* and *Pirates of the Caribbean*.

Vision-based interaction includes camera-based tracking for games and interfaces. Examples include infrared tracking with the Nintendo Wii and camera tracking of crowds for interactive displays.

6.6 Vehicles, Robotics, and Space Exploration

Applications include autonomous cars, planes, missiles, robots, space exploration, and manufacturing or quality control. Smart-car systems use vision for road-scene interpretation.

A historical automotive example cited Mobileye systems in high-end BMW, GM, and Volvo models and projected broad manufacturer adoption by 2010.

Mars exploration provides a detailed robotics example. Vision systems at JPL were used for panorama stitching, three-dimensional terrain modeling, obstacle detection, and position tracking. Robotics examples also include the Mars Spirit Rover and RoboCup.

6.7 Medical and Scientific Imaging

Medical imaging includes three-dimensional imaging with modalities such as MRI and CT, as well as image-guided surgery.

6.8 Image Retrieval and Three-Dimensional Viewing

Image retrieval supports finding and browsing pictures in large collections. Three-dimensional Earth viewers are examples of large-scale three-dimensional modeling and visualization.

6.9 Application Overview

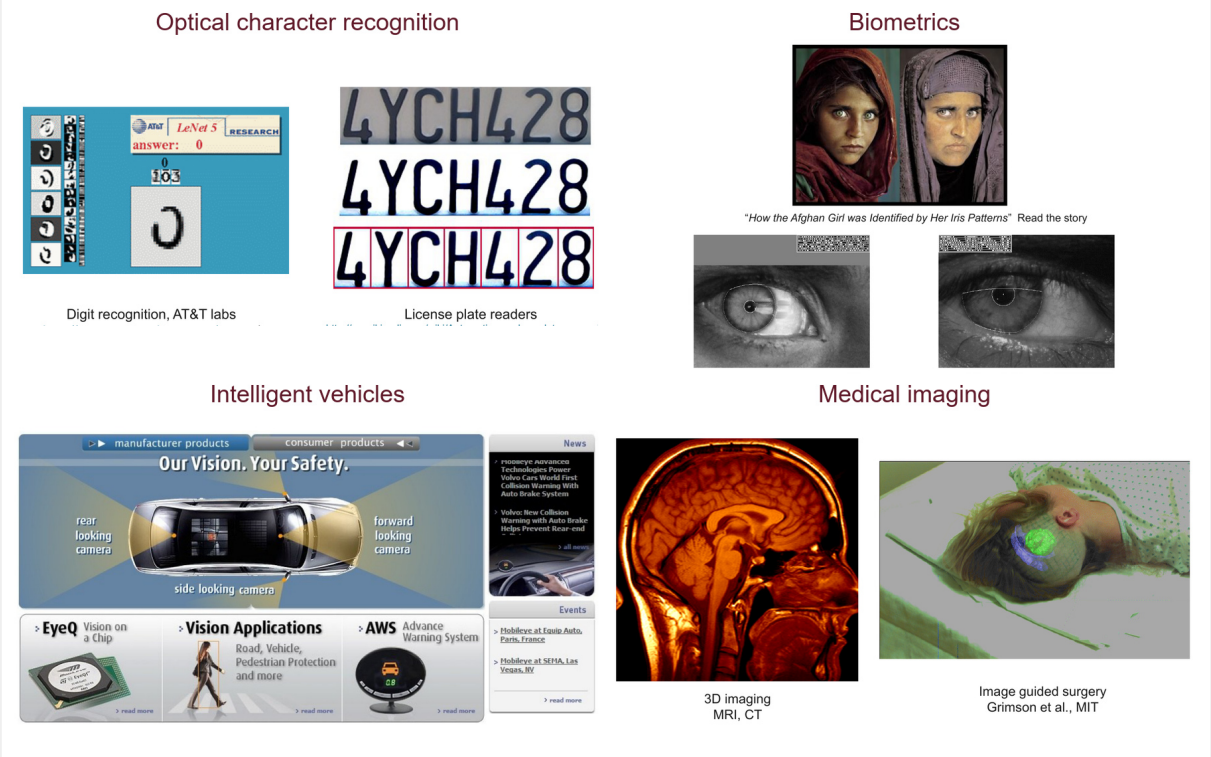


Figure 6.2. Examples of computer-vision applications in OCR, biometrics, intelligent vehicles, and medical imaging.

Table 6.1. Representative computer-vision application areas.

Application area	Representative examples
Transportation and autonomy	smart cars, autonomous vehicles, planes, robots
Space exploration	panorama stitching, 3D terrain modeling, obstacle detection, tracking
Accessibility	aid to the blind, ASL recognition
Manufacturing	quality control, retail product recognition
Security and identity	surveillance, biometrics, face / iris / fingerprint recognition
Documents	OCR, digit recognition, license-plate reading
Medical imaging	MRI, CT, 3D imaging, image-guided surgery
Entertainment and interaction	special effects, motion capture, camera-based interaction
Image collections	retrieval, browsing, image-based rendering

Chapter summary. Computer vision supports recognition, documents, retail, biometrics, special effects, interaction, autonomous systems, robotics, space exploration, medical imaging, and image retrieval. These applications combine different subsets of sensing, geometry, grouping, motion, and recognition.

CHAPTER 7

Computer Vision and Machine Learning

Machine learning is an important tool for computer vision, but the two fields are not identical. Computer vision also includes sensing, image formation, geometric modeling, and visual representation.

7.1 The Role of Machine Learning

Machine learning studies algorithms that can change behavior based on data. A major goal is to learn complex patterns and make decisions from data. This is directly useful in computer vision, especially for recognition.

Computer vision, however, is more than learning. It also includes explicit modeling and example-based learning, and it cares about how visual data are obtained and represented.

Table 7.1. Contrast between the emphases described for machine learning and computer vision.

	Machine learning emphasis	Computer vision emphasis
Data	Learns patterns and decisions from data	Also considers how visual data are acquired
Sensors	Often treats the data source as given	Includes sensor design and active vision
Representation	Learns or uses representations for prediction	Explicitly studies visual representation
Modeling	Data-driven methods are central	Modeling and learning both matter

The source's pattern-recognition example separates **training** from **testing**. Selected measurements (features) from training images are used to train a statistical method; corresponding measurements from testing images are then used to predict a category. Representative methods shown are support vector machines (SVM), Gaussian Naive Bayes (GNB), and neural networks.



Figure 7.1. Training and testing in the lecture's statistical pattern-recognition example.

7.2 Related Disciplines

Computer vision is connected to image processing, pattern recognition, photogrammetry, computer graphics, artificial intelligence, machine learning, projective geometry, and control theory.

7.3 Active Research Topics

Representative research topics include object recognition, human behavior analysis, continuous computer-vision streams, biometrics and soft biometrics, large-scale three-dimensional reconstruction at city scale, medical image processing, and vision for robotics.

7.4 Research Publications

Major publication venues include *IEEE Transactions on Pattern Analysis and Machine Intelligence* (TPAMI), the *International Journal of Computer Vision* (IJCV), and *IEEE Transactions on Image Processing*. Major conferences include ICCV, CVPR, and ECCV.

> **Historical publication snapshot**
>
> A historical publication snapshot lists TPAMI with an impact factor of 17.730 and IJCV with 11.541. These values are historical rather than current metrics. CVPR is described as annual, while ICCV and ECCV are described as occurring every two years.

The publication rankings are used to emphasize the importance of computer vision research within computer science and electrical engineering.

7.5 Further Reading

Recommended and optional books include:

- David Forsyth and Jean Ponce, *Computer Vision: A Modern Approach*;
- Richard Szeliski, *Computer Vision: Algorithms and Applications*;
- Emanuele Trucco and Alessandro Verri, *Introductory Techniques for 3-D Computer Vision*;
- Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*.

> **Chapter summary.** Machine learning provides important methods for visual recognition, but computer vision additionally studies cameras, sensors, image formation, geometry, representation, and visual data acquisition. The field connects to several neighboring disciplines and remains active across recognition, reconstruction, biometrics, medicine, and robotics.

Lecture 1 Quick Review

Foundations and visual cues

- **Purpose**: extract geometric, spatial, and semantic properties of the 3D world from images; graphics follows the synthesis direction.
- **Ambiguity**: 3D-to-2D projection is ill-posed. Illumination, pose, occlusion, viewpoint, scale, deformation, clutter, and intra-class variation affect appearance; context helps interpretation.
- **Depth and shape**: linear/aerial perspective, occlusion, texture gradients, shading, and cast shadows provide different cues.
- **Grouping and scale**: similarity and common fate help organization; millions of pixels and many categories make vision computationally demanding.

History and understanding

- Development moved through structured representations, geometry, statistics, annotated datasets, and deep learning.
- **Measurement** includes stereo, structure from motion, and multi-view reconstruction; **semantics** includes objects, scenes, and qualitative spatial relations.

Core areas and methods

- **Early vision**: image formation, cameras, light, color, filtering, edges, and local patches. Shifted filters compare image patches with patterns.
- **Texture**: filter-response statistics support segmentation and shape interpretation; patches also support correspondence.
- **Mid-level and multi-view**: fitting, grouping, segmentation, tracking, stereo, and projective structure from motion.
- **Recognition**: patch descriptions, visual vocabularies, bag-of-features, and classification. Higher geometric vision studies models, surfaces, outlines, aspect graphs, and range data.

Applications and learning

- OCR, biometrics, medicine, autonomous systems, robotics, space, effects, interaction, and retrieval combine sensing, geometry, and recognition.
- Face detection can control autofocus, exposure, and flash. Rover vision supports panoramas, terrain models, obstacles, and position tracking.
- **Machine learning** learns patterns from data; vision additionally studies sensing, acquisition, modeling, and representation.
- Related fields include image processing, pattern recognition, photogrammetry, graphics, AI, geometry, learning, and control.
- Research includes recognition, behavior, biometrics, reconstruction, medicine, and robotics. Major venues include TPAMI, IJCV, IEEE TIP, CVPR, ICCV, and ECCV.

LECTURE 2

Image Acquisition and Digitalization

From scene illumination to a sampled digital image.

Image formation · Digitalization · Pixel relations · Interpolation

Computer Vision — Marco Raoul Marini

Notes prepared by Matt Clifford Trinidad Ramos

CHAPTER 8

Image Formation and Camera Acquisition

The acquisition lecture follows the physical path from illumination to a digital image. A scene point receives incident light, the surface modifies that light according to its reflectance and other optical properties, the reflected light is projected onto the image plane, and the sensor converts wavelength-dependent energy into one or more measured image channels.

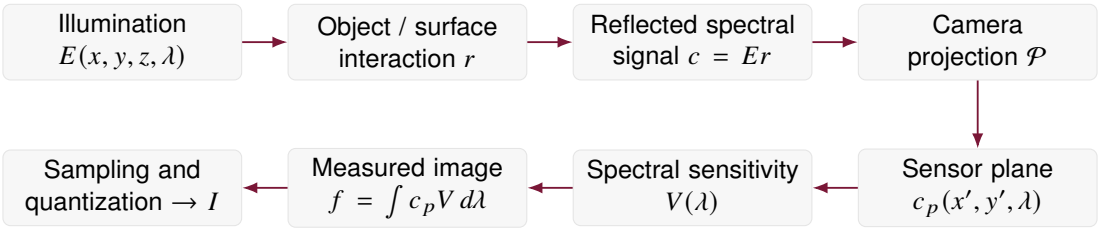


Figure 8.1. Acquisition chain connecting the lecture's physical, geometric, spectral, and digital stages. Symbols are defined in the following equations; I is the digital image.

8.1 Illumination and Surface Properties

The slides contrast typical source strengths and surface reflectivities to emphasize that observed brightness depends on both illumination and the object. Representative values shown in the lecture are collected in Table 8.1.

Table 8.1. Representative illumination and reflectivity values reported in the lecture.

Light source	i (candela)	Object / surface	r
Solar light	9000	Black velvet	0.01
Cloudy sky	1000	White wall	0.80
Moon light	0.01	Silver and other metals	0.90
Internal office light	100	Snow	0.93

The lecture distinguishes five surface interactions: **absorption**, **diffusion**, **reflection**, **transparency**, and **refraction**. They describe different ways in which incoming light can be attenuated, scattered, redirected, transmitted, or bent at an interface. The wavelength λ matters because both illumination and material response can vary spectrally.

8.2 A Spectral Image-Formation Model

For a natural image, let $E(x, y, z, \lambda)$ denote the incident light at a world point and let $r(x, y, z, \lambda)$ denote the point's reflectivity. The reflected light is modeled in the lecture as

$$c(x, y, z, \lambda) = E(x, y, z, \lambda)\, r(x, y, z, \lambda). \tag{8.1}$$

where

- $E(x, y, z, \lambda)$: incident spectral light at world location (x, y, z);
- $r(x, y, z, \lambda)$: wavelength-dependent surface reflectivity;
- $c(x, y, z, \lambda)$: reflected spectral light leaving that point toward the imaging system;
- λ: wavelength.

> **From scene radiance to an image**
>
> The camera does not directly store the three-dimensional spectral function. It first projects world coordinates into two-dimensional image coordinates and then applies the wavelength sensitivity of the capture device.

The projection operator $\mathcal{P}$ maps a world point to camera coordinates:

$$c_p(x', y', \lambda) = \mathcal{P}(c(x, y, z, \lambda)) \, . \tag{8.2}$$

where

- (x, y, z): world coordinates;
- (x', y'): image-plane or camera coordinates;
- c_p: projected spectral signal at the sensor plane;
- $\mathcal{P}$: camera projection operator.

Once the projected signal reaches the sensor, the device sensitivity $V(\lambda)$ determines how much of each wavelength contributes to the measured image value:

$$f(x', y') = \int c_p(x', y', \lambda) V(\lambda) \, d\lambda. \tag{8.3}$$

where

- $V(\lambda)$: spectral sensitivity function of the capture device;
- $f(x', y')$: scalar image function measured at image location (x', y');
- the integral: accumulation of wavelength-dependent energy weighted by sensor sensitivity.

8.3 Perspective and Orthographic Projection

The lecture compares two projection models. In a **perspective projection**, objects closer to the camera appear larger; this matches the human eye and ordinary cameras and is therefore natural, but mathematically more involved. In an **orthographic projection**, apparent object size is independent of distance; this is less realistic but simpler to analyze.

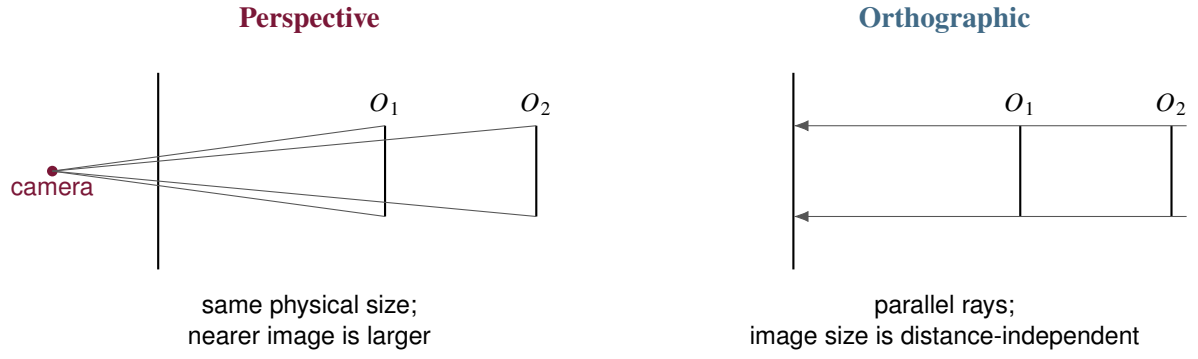


Figure 8.2. Conceptual contrast between perspective and orthographic projection.

8.4 Color Sensors

A color camera can be modeled as three wavelength-selective measurements at each image location. With red, green, and blue sensitivity functions V_R, V_G, and V_B, the lecture writes

$$\begin{aligned} f_R(x', y') &= \int c_p(x', y', \lambda) V_R(\lambda)\, d\lambda, \\ f_G(x', y') &= \int c_p(x', y', \lambda) V_G(\lambda)\, d\lambda, \\ f_B(x', y') &= \int c_p(x', y', \lambda) V_B(\lambda)\, d\lambda. \end{aligned} \tag{8.4}$$

where

- f_R, f_G, f_B: the three measured image channels;
- V_R, V_G, V_B: red-, green-, and blue-tuned sensor sensitivity functions;
- c_p: projected spectral signal reaching the sensor.

The three channel values are then sufficient for ordinary RGB display devices to reconstruct a color appearance, even though the original spectrum contains much more information.

> **Chapter summary.** Image formation is a chain: illumination → surface interaction → reflected spectral light → geometric projection → sensor sensitivity. Perspective and orthographic projection differ in whether apparent size changes with depth, while color acquisition uses multiple spectral sensitivity functions rather than one scalar response.

CHAPTER 9

Digital Images: Sampling, Quantization, and Resolution

9.1 From a Continuous Image to a Digital Image

Before digitalization, the projected image is continuous in both its spatial domain and its value range: $(x', y') \in \mathbb{R}^2$ and $f(x', y') \in \mathbb{R}$. A digital computer therefore needs two discretization steps.

> **Sampling and quantization**
>
> **Sampling** discretizes spatial coordinates, producing a finite grid of locations. **Quantization** discretizes the value measured at each sampled location, producing one of a finite set of representable levels.

The lecture expresses the two operations schematically as

$$(x', y') \longrightarrow (x'_i, y'_j), \qquad i = 0, \ldots, N-1, \quad j = 0, \ldots, M-1, \tag{9.1}$$

$$f(x'_i, y'_j) \longrightarrow \hat{f}(x'_i, y'_j) \in \mathbb{N}. \tag{9.2}$$

where

- N, M: numbers of sample positions along the two image axes;
- (x'_i, y'_j): discrete sample location;
- f: continuous-valued image before quantization;
- $\hat{f}$: quantized digital value.

The sampling slides represent spatial sampling as multiplication by a finite **comb function**:

$$\text{comb}(x', y') = \sum_{i=0}^{N-1} \sum_{j=0}^{M-1} \delta(x' - i\Delta_x,\ y' - j\Delta_y). \tag{9.3}$$

where

- δ: two-dimensional Dirac impulse selecting a grid location;
- Δ_x, Δ_y: spacing between samples along the two axes;
- i, j: grid indices, with N horizontal and M vertical sample positions;
- x', y': continuous image-plane coordinates.

The slide writes the operation schematically as

$$f(x'_i, y'_j) = f(x', y') \times \text{comb}(x', y'), \qquad f(x'_i, y'_j) = f(i, j) \text{ for simplicity.} \tag{9.4}$$

Here $x'_i = i\Delta_x$ and $y'_j = j\Delta_y$; $f(i, j)$ denotes the value at that sample, before quantization. The product on the right is an impulse representation of the sampled image; its impulse weights are the sample values on the left. This distinction makes the slide's shorthand precise without changing its sampling construction.

A digital image is therefore a multidimensional array. For a monochrome image, a pixel stores one intensity value; for a color image, it stores a vector of channel values. The lecture also notes other coordinate structures: (x, y, z) for volumetric images such as CT scans and (x, y, t) for video.

9.2 Quantization and Bit Depth

If each pixel value is represented by b bits, the number of available quantization levels is

$$N_c = 2^b. \tag{9.5}$$

where

- N_c: number of representable gray levels or colors;
- b: number of bits used for each quantized value.

For example, $b = 8$ gives $2^8 = 256$ levels. Reducing the number of levels creates visible **false contours** because smooth intensity changes collapse into coarse steps.

After sampling and quantization, the lecture describes an image as a discrete function with a finite discrete domain and range:

$$\Omega \subset \mathbb{N}^2, \qquad I : \Omega \longrightarrow \{1, \ldots, P\}, \qquad P \in \mathbb{N}. \tag{9.6}$$

where

- Ω: finite set of pixel coordinates selected by sampling;
- I: image assigning a quantized value to each pixel;
- P: number of representable levels; the source's 8-bit example uses $P = 2^8 = 256$ and $\log_2 P = 8$.

The slide numbers the levels from 1 to P; this expresses the number of levels, while binary image values elsewhere use 0 and 1.

Digital image processing means processing digital, multidimensional images using a digital computer.

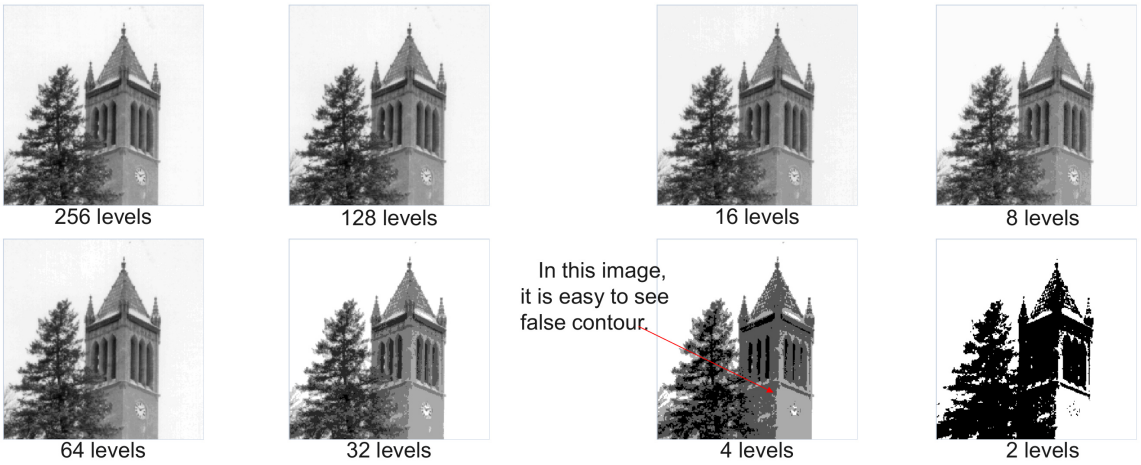


Figure 9.1. Source examples of progressively coarser quantization; false contouring becomes evident at very low bit depth.

9.3 Digital Image Types

Table 9.1. Digital image representations presented in the lecture.

Type	Pixel content	Interpretation
Intensity / monochrome	one scalar	light intensity, commonly displayed as gray level
RGB color	a 3-component vector	red, green, and blue channel values
Binary	one bit	0 for black and 1 for white in the lecture convention
Indexed	one integer index	lookup into a color table containing RGB triples

9.4 Pixels, Memory, and Spatial Resolution

A **pixel** is a single value on the sampled grid. The lecture stresses that a pixel should be understood as the smallest treatable image area, not as an infinitesimal geometric point.

For an $M \times N$ image with L intensity levels, binary storage uses

$$k = \log_2 L, \qquad B = MNk, \tag{9.7}$$

where

- M, N: image dimensions in pixels;
- L: number of intensity levels;
- k: bits per pixel value;
- B: total number of bits for the image.

> **Notation in the source**
>
> One slide writes $k = \log(L)$, while the later 8-bit example explicitly uses $\log_2(P) = 8$ for $P = 256$. The binary-storage interpretation therefore uses base-2 logarithms.

Spatial resolution is the smallest distinguishable detail in an image. The lecture lists pixels per inch (ppi), dots per inch (dpi), and raw $M \times N$ dimensions as common descriptors, while warning that pixel dimensions alone do not fully express the relation between digital sampling and the detail present in the original scene.

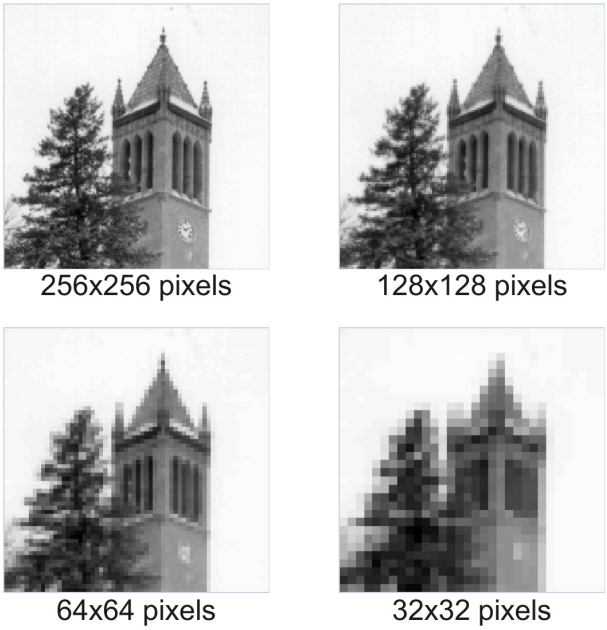


Figure 9.2. Effect of reducing spatial sampling while preserving the same field of view: fine detail is progressively lost.

Under-sampling is irreversible: interpolation can produce a denser grid, but it cannot recover scene details that were never sampled. The lecture gives application-dependent resolution examples, such as monitor/web display around 72 dpi and printing at substantially higher dpi, and presents interpolation as an approximation when increasing image size.

Resolution and perceived quality

For images displayed at the same physical size, insufficient spatial sampling produces visible pixels and lost detail. If the pixel dimensions are kept fixed instead, changing the number of samples also changes the displayed image size. The suitable combination of image size and pixel depth is application- and content-dependent.

Chapter summary. Digitalization separates two independent limitations: spatial sampling controls where the image is measured, while quantization controls how finely each measured value is represented. Bit depth determines the number of levels; spatial resolution determines recoverable detail; interpolation only estimates values between existing samples.

CHAPTER 10

Pixel Relationships and Image Resampling

10.1 Neighborhoods

For a pixel $p = (x, y)$, the lecture defines the four-neighborhood, diagonal neighborhood, and eight-neighborhood:

$$N_4(p) = \{(x-1, y), (x+1, y), (x, y-1), (x, y+1)\}, \tag{10.1}$$

$$N_D(p) = \{(x-1, y-1), (x+1, y-1), (x-1, y+1), (x+1, y+1)\}, \tag{10.2}$$

$$N_8(p) = N_4(p) \cup N_D(p). \tag{10.3}$$

where

- $p = (x, y)$: reference pixel;
- $N_4(p)$: horizontal and vertical neighbors;
- $N_D(p)$: four diagonal neighbors;
- $N_8(p)$: all eight surrounding grid locations.

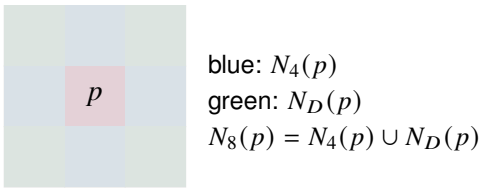


Figure 10.1. Pixel-neighborhood relations used to define connectivity, adjacency, and paths.

10.2 Connectivity, Adjacency, and Paths

Two pixels from the same class are connected when the selected neighborhood relation links them. The lecture distinguishes **4-connectivity**, **8-connectivity**, and **mixed (m-) connectivity**. Let V be the set of pixel values defining the selected class. For pixels p, q with values in V, m-adjacency holds if either $q \in N_4(p)$, or $q \in N_D(p)$ and no pixel in $N_4(p) \cap N_4(q)$ has a value in V. Connectivity means that a path of adjacent pixels exists; the adjacency rule itself describes a single step. The mixed rule is introduced to avoid the ambiguity that diagonal links can create under pure 8-connectivity.

> **Restrict the rule to the selected class**
>
> The source's written rule omits the restriction to the selected pixel class. Diagonal pixels on an interior grid always share two 4-neighbors; m-adjacency requires that neither shared neighbor belongs to the selected class. In this diagram, the dashed diagonal is excluded because it shares a selected neighbor, whereas the final diagonal to q is allowed because its shared neighbors are outside the class.

A pixel p is **adjacent** to q when the selected local neighborhood rule links them. Two subsets S_1 and S_2 are adjacent if some pixel in one subset is adjacent to a pixel in the other. A path from $p = (x, y)$ to

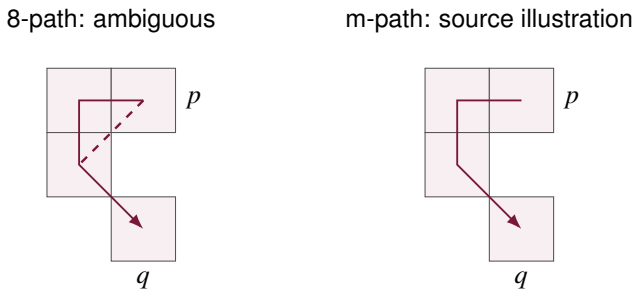


Figure 10.2. The source example removes the alternative diagonal route in its m-path illustration, retaining a single route from p to q.

$q = (s, t)$ is a sequence of distinct adjacent pixels $(x_0, y_0), \ldots, (x_n, y_n)$ with the required endpoints; the adjacency convention determines whether the result is a 4-path, 8-path, or m-path.

10.3 Distance Measures

A distance D must satisfy non-negativity, identity of indiscernibles, symmetry, and the triangle inequality. Three distances appear in the lecture:

$$D_E(p, q) = \sqrt{(x - s)^2 + (y - t)^2}, \tag{10.4}$$

$$D_4(p, q) = |x - s| + |y - t|, \tag{10.5}$$

$$D_8(p, q) = \max\{|x - s|, |y - t|\}. \tag{10.6}$$

where

- $p = (x, y)$ and $q = (s, t)$: two pixel coordinates;
- D_E: Euclidean distance;
- D_4: city-block distance, whose unit-distance pixels are exactly the 4-neighbors;
- D_8: chessboard distance, whose unit-distance pixels are exactly the 8-neighbors.

10.4 Interpolation

Interpolation estimates the value of an image at non-integer coordinates. The lecture separates resizing into two steps:

1. **Create the new sampling grid**, mapping its positions into the original image coordinates. The slide's $1.5\times$ zoom changes a 500×500 grid to 750×750.
2. **Assign a value to each new pixel** by interpolation of the existing samples.

The lecture compares four approaches.

Nearest-neighbor interpolation is written in the slides as

$$f(x, y, z) = \mathrm{Im}(\mathrm{round}(x), \mathrm{round}(y), z), \tag{10.7}$$

Here $f(x, y, z)$ is the interpolated image value, Im is the original sampled image, x, y are source spatial coordinates, round selects the nearest integer, and z remains an integer channel index in the implementation example.

For triangle interpolation, the slide uses the schematic weighted sum

$$Q = V_1 A_1 + V_2 A_2 + V_3 A_3, \tag{10.8}$$

Table 10.1. Interpolation methods discussed in the acquisition lecture.

Method	Principle	Lecture emphasis
Nearest neighbor	choose the closest sampled pixel	fastest; blocky appearance; be careful with integer rounding
Triangle interpolation	weighted combination of three triangle vertices	useful for less-structured samples; weights follow opposite sub-areas and should be normalized
Bilinear	weighted combination of the four corners of a grid cell	smoother than nearest neighbor; four lookups; often a good speed/quality tradeoff
Bicubic	cubic interpolation using a 4×4 neighborhood	16 neighbors; smoother than bilinear; more computation

Here Q is the value assigned at the interior point q, V_i are vertex values, and A_i are opposite sub-triangle areas, for $i = 1, 2, 3$. This is the **slide's schematic expression**. The slide explicitly requires normalization by total area in the general case. Its **normalized interpretation** is

$$Q = \frac{V_1A_1 + V_2A_2 + V_3A_3}{A_1 + A_2 + A_3}. \tag{10.9}$$

The denominator is the whole triangle's area; the weights $A_i/(A_1 + A_2 + A_3)$ therefore sum to one.

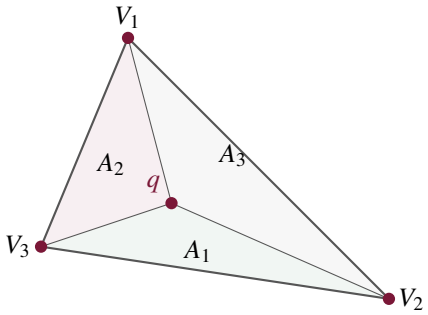


Figure 10.3. Triangle construction: A_i is the sub-triangle opposite vertex V_i. Normalize by $A_1 + A_2 + A_3$ to obtain area weights.

For bilinear interpolation inside a unit grid cell, the four opposite rectangle areas are

$$A_1 = d_2d_4, \quad A_2 = d_1d_4, \quad A_3 = d_2d_3, \quad A_4 = d_1d_3, \tag{10.10}$$

and the interpolated value is

$$Q = V_1A_1 + V_2A_2 + V_3A_3 + V_4A_4. \tag{10.11}$$

where

- Q: interpolated value at the position q; $V_1, \ldots, V_4$: values at the four surrounding grid points;
- d_1, d_2, d_3, d_4: horizontal and vertical fractional distances shown in the lecture construction;
- A_i: area opposite vertex V_i; within a unit cell the weights sum to one.

Bicubic interpolation is presented as a cubic interpolation of four cubic interpolations. It uses the 16 nearest samples, fits third-order polynomials such as $f(x) = a + bx + cx^2 + dx^3$, interpolates along one axis, and then fits/interpolates along the other. Here x is the interpolation coordinate, $f(x)$ is its estimated value, and a, b, c, d are the fitted polynomial coefficients.

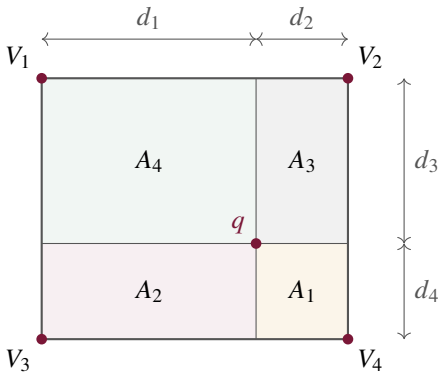


Figure 10.4. Bilinear geometry redrawn from the source: $d_1 + d_2 = d_3 + d_4 = 1$ in the unit cell. Each corner uses the opposite rectangle's area; Q is the value at q. The drawing is enlarged for readability.



Figure 10.5. Lecture comparison of bilinear and bicubic resampling on the same image region.

The resize algorithm is consequently simple at a high level: for every pixel in the new image, map its position back into the original-image coordinates, interpolate the source value, and write that value into the output image.

> **Chapter summary.** Neighborhood relations define connectivity, adjacency, paths, and grid-aware distance measures. Resampling then estimates values between grid points: nearest neighbor is cheapest, bilinear provides a common compromise, and bicubic trades additional computation for smoother results. Increasing sample density by interpolation does not reverse information lost during down-sampling.

Lecture 2 Quick Review

Physical acquisition

1. Reflected light is modeled by incident illumination multiplied by surface reflectivity.
2. Projection maps the 3D scene into 2D image coordinates.
3. Sensor sensitivity integrates the projected spectrum over wavelength.
4. RGB acquisition uses three different spectral sensitivity functions.
5. Perspective projection changes apparent size with depth; orthographic projection does not.

Digitalization and storage

6. Sampling discretizes position; quantization discretizes value.
7. b bits provide 2^b representable levels.
8. Low quantization depth can create false contours.
9. A pixel is the smallest treatable sampled image area.
10. For $M \times N$ pixels and k bits per value, storage is MNk bits.

Pixel relations and interpolation

11. N_4, N_D, and N_8 define local grid neighborhoods.
12. m-connectivity avoids some diagonal ambiguity of 8-connectivity.
13. D_4 is city-block distance; D_8 is chessboard distance.
14. Nearest neighbor is fast but blocky; bilinear uses four samples; bicubic uses sixteen.
15. Interpolation creates estimates, not lost high-frequency detail.

LECTURE 3

Color, Perception, and Texture

From spectral measurements to color and appearance.

Human vision · Color spaces · White balance · Texture

Computer Vision — Marco Raoul Marini

Notes prepared by Matt Clifford Trinidad Ramos

CHAPTER 11

Color: Light and Human Vision

11.1 The Nature of Color

The lecture emphasizes that color is not simply a physical property attached to an object. It is a **psychological property of visual experience** produced by the interaction between the spectrum of physical light and the human visual system. This distinction motivates both colorimetry and color constancy: a camera measures spectral effects, whereas a human observer experiences a perceptual color.

Visible light occupies only a limited interval of the electromagnetic spectrum, approximately 400–700 nm in the lecture. A light source can be described physically by its spectral power distribution: the amount of energy emitted at each wavelength. The lecture contrasts narrow-band sources such as a ruby laser with broader spectra such as tungsten illumination and daylight.

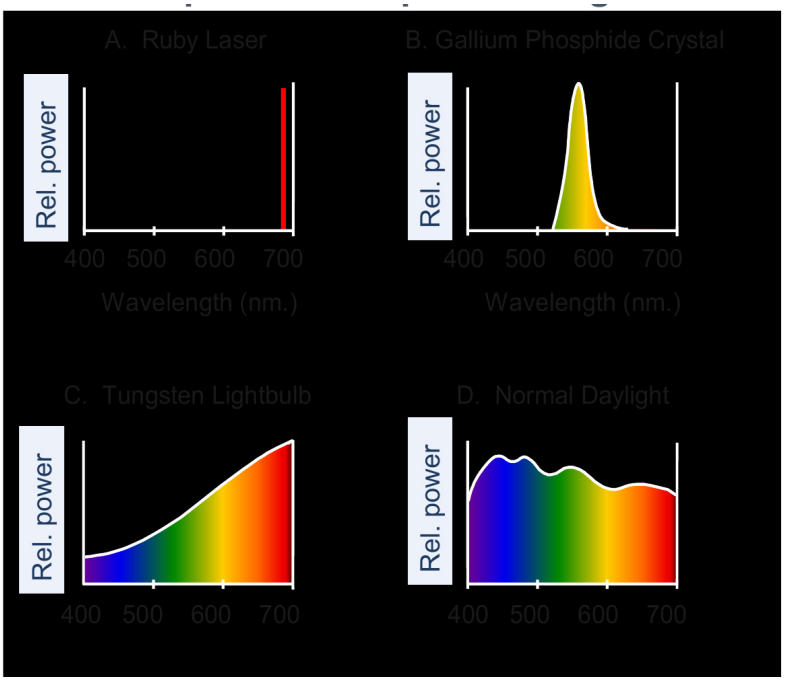


Figure 11.1. Examples from the lecture of qualitatively different source spectra, including narrow-band and broadband illumination.

A surface is similarly characterized by its reflectance spectrum: the fraction of incident light reflected at each wavelength. Therefore the observed color signal depends jointly on source spectrum and surface reflectance.

11.2 The Eye as a Sensing System

The lecture draws a camera analogy for the eye: the iris regulates the pupil aperture, the lens changes shape to focus at different distances, and the retina supplies the photoreceptors. Rods and cones are not uniformly distributed. Rods are emphasized for intensity sensitivity, while cones support color vision. The **fovea**, a

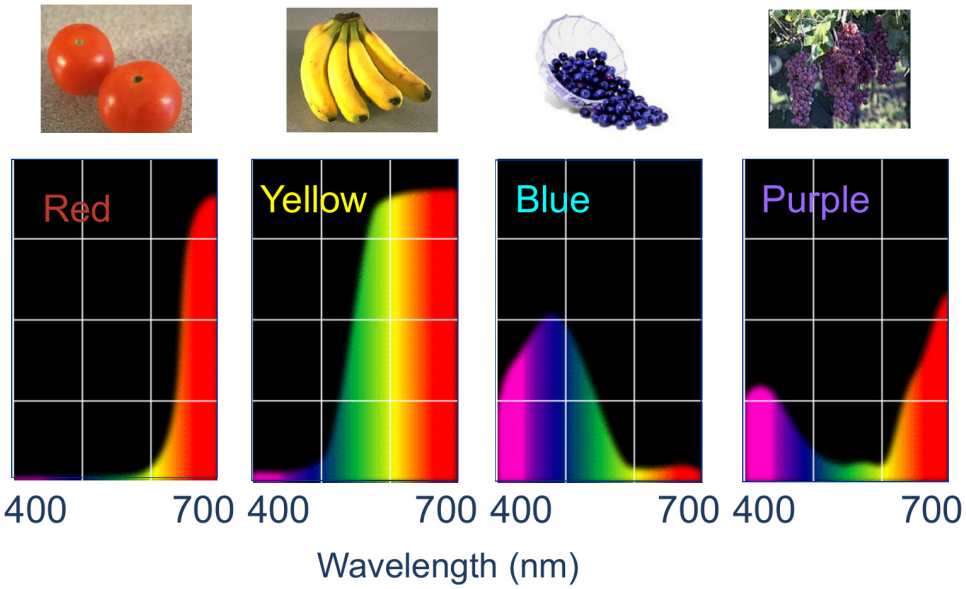


Figure 11.2. Representative surface reflectance spectra shown in the lecture for red, yellow, blue, and purple surfaces.

small central region covering roughly 1–2° of visual angle in the slides, has the highest cone density and no rods; peripheral acuity is lower because many rods can feed the same neuron.

Human color vision uses three cone classes, conventionally called S, M, and L for their relative wavelength sensitivities. The lecture gives an approximate L:M:S ratio of 10 : 5 : 1 and notes that there are almost no S cones in the center of the fovea.

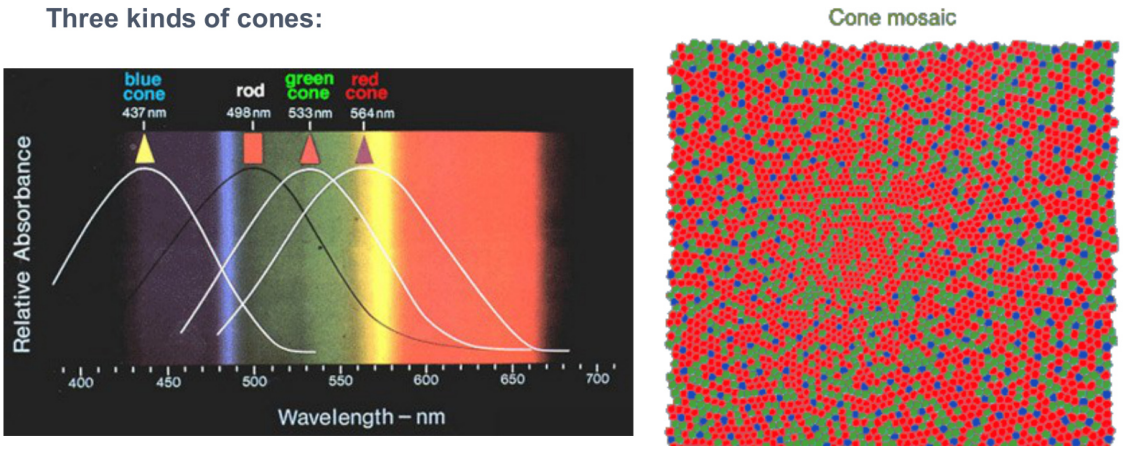


Figure 11.3. Lecture illustration of the three cone-sensitivity functions and their non-uniform representation.

Rods operate at lower illumination than cones. Below the color-perception threshold, rod-dominated vision can still detect brightness, while cone-based color and fine foveal detail become ineffective. This connects the sensitivity slide's question, "Why can't we read in the dark?", to the absence of rods in the fovea and the lower acuity of peripheral rod vision. The source chart distinguishes the threshold in bright sunlight from the much lower threshold after dark adaptation. Light/dark adaptation changes sensitivity as illumination changes; it is a gradual effect, unlike instantaneous contrast.

Brewster's-color demonstrations are used as evidence that the visual system can interpolate color from spatially offset samples. This is consistent with the fact that the three cone populations do not sample every retinal location identically.

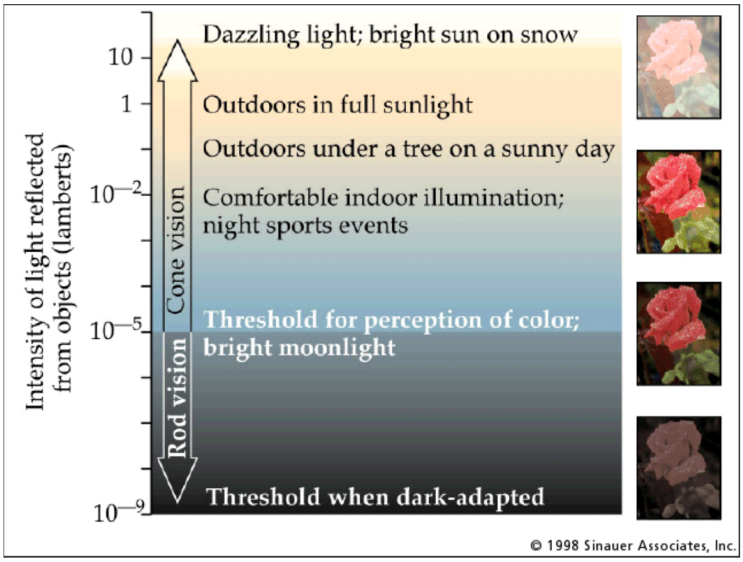


Figure 11.4. Source sensitivity regimes and light/dark adaptation thresholds. The chart is preserved rather than assigning invented numerical sensitivity curves.

11.3 Spectral Filtering and Metamers

A cone response can be understood as a spectral filter: multiply the incoming spectrum by the receptor sensitivity curve and integrate over wavelength. Each cone class therefore compresses an entire spectrum into a single response. Three cone outputs cannot preserve the full spectral signal, so distinct spectra can produce identical responses.

Metamerism

Metamers are physically different spectral power distributions that produce the same perceived color under the relevant viewing conditions. Metamerism is a direct consequence of compressing a high-dimensional spectrum into a small number of receptor responses.

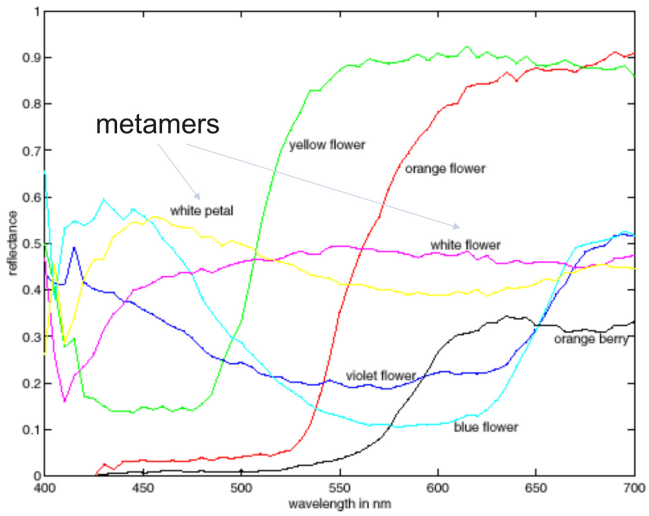


Figure 11.5. Real-world spectral curves used in the lecture to illustrate that different spectra may be perceptually matched.

Chapter summary. Physical spectra are high-dimensional, but human color perception is driven mainly by three cone classes. Consequently, color depends on both illumination and surface reflectance, and different spectra can become perceptually indistinguishable as metamers.

CHAPTER 12

Color Matching and Color Spaces

12.1 Trichromacy and Color Matching

Color-matching experiments standardize which spectra produce the same sensation under controlled conditions. The lecture's central empirical result is **trichromacy**: for most observers, three primaries are sufficient to match a wide range of test lights, with color blindness noted as an exception. The same test light and same primaries tend to produce similar matching weights across observers.

In the experiment, the observer compares a test light with a mixture of three primary lights and adjusts their weights until the two fields appear to match.

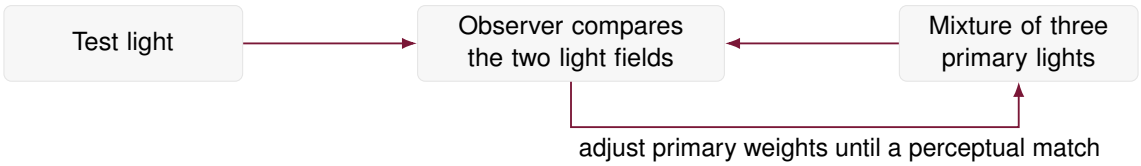


Figure 12.1. Conceptual color-matching experiment underlying trichromacy and Grassmann's laws.

12.2 Grassmann's Laws

The lecture states that additive color matching is linear. If

$$A = u_1P_1 + u_2P_2 + u_3P_3,$$
$$B = v_1P_1 + v_2P_2 + v_3P_3,$$

then mixtures and scalings obey

$$A + B = (u_1 + v_1)P_1 + (u_2 + v_2)P_2 + (u_3 + v_3)P_3, \tag{12.1}$$
$$kA = (ku_1)P_1 + (ku_2)P_2 + (ku_3)P_3. \tag{12.2}$$

where

- P_1, P_2, P_3: chosen primary lights;
- u_i, v_i: primary weights that match test lights A and B;
- k: scalar intensity multiplier.

If two test lights are matched by the same three weights, the color-matching experiment treats them as a perceptual match.

12.3 Color-Matching Functions as a Linear Transform

Choose three primaries and measure the weights $c_1(\lambda)$, $c_2(\lambda)$, and $c_3(\lambda)$ required to match monochromatic lights at sampled wavelengths. Store these color-matching functions as the rows of a matrix

$$C = \begin{bmatrix} c_1(\lambda_1) & \cdots & c_1(\lambda_N) \\ c_2(\lambda_1) & \cdots & c_2(\lambda_N) \\ c_3(\lambda_1) & \cdots & c_3(\lambda_N) \end{bmatrix}. \tag{12.3}$$

Represent a new sampled spectral signal by

$$\mathbf{t} = \begin{bmatrix} t(\lambda_1) & \cdots & t(\lambda_N) \end{bmatrix}^{\mathsf{T}}.$$

The primary coordinates are then

$$\mathbf{e} = C\mathbf{t}. \tag{12.4}$$

where

- C: $3 \times N$ matrix of sampled color-matching functions;
- $\mathbf{t}$: sampled spectral power distribution of the test light;
- $\mathbf{e} = (e_1, e_2, e_3)^{\mathsf{T}}$: three primary weights that describe its color in the chosen linear color space; $\lambda_1, \ldots, \lambda_N$ are the N sampled wavelengths, $t(\lambda_i)$ is spectral power, and $c_j(\lambda_i)$ is the matching weight for primary j.

12.4 RGB and CIE XYZ

A **linear color space** is defined by a choice of three primaries. Mixing two lights traces a straight line between their coordinates; mixtures of three primaries lie within the triangle they define. In RGB, the primaries correspond to red, green, and blue lights. The source figure specifies $p_1 = 645.2\,\text{nm}$ (red), $p_2 = 525.3\,\text{nm}$ (green), and $p_3 = 444.4\,\text{nm}$ (blue); these are the selected monochromatic primaries in that lecture example. The lecture notes that some wavelengths require subtractive matching, which appears as negative values in RGB color-matching functions.

The CIE XYZ system, standardized in 1931 by the International Commission on Illumination, uses imaginary primaries chosen so that its matching functions are nonnegative. A two-dimensional chromaticity representation is obtained by normalizing:

$$x = \frac{X}{X+Y+Z}, \qquad y = \frac{Y}{X+Y+Z}. \tag{12.5}$$

where

- X, Y, Z: CIE tristimulus coordinates;
- x, y: normalized chromaticity coordinates that remove the common scale $X + Y + Z$ (defined when this sum is nonzero).

12.5 Toward Perceptual Uniformity

The lecture stresses that Euclidean differences in CIE x, y coordinates are not perceptually uniform. MacAdam ellipses illustrate that just-noticeable color differences occupy differently sized and oriented regions of the chromaticity diagram. CIE $u'v'$ is introduced as a projective transform intended to make these regions more uniform, followed by CIE $L^*a^*b^*$ as a later perceptually motivated space.

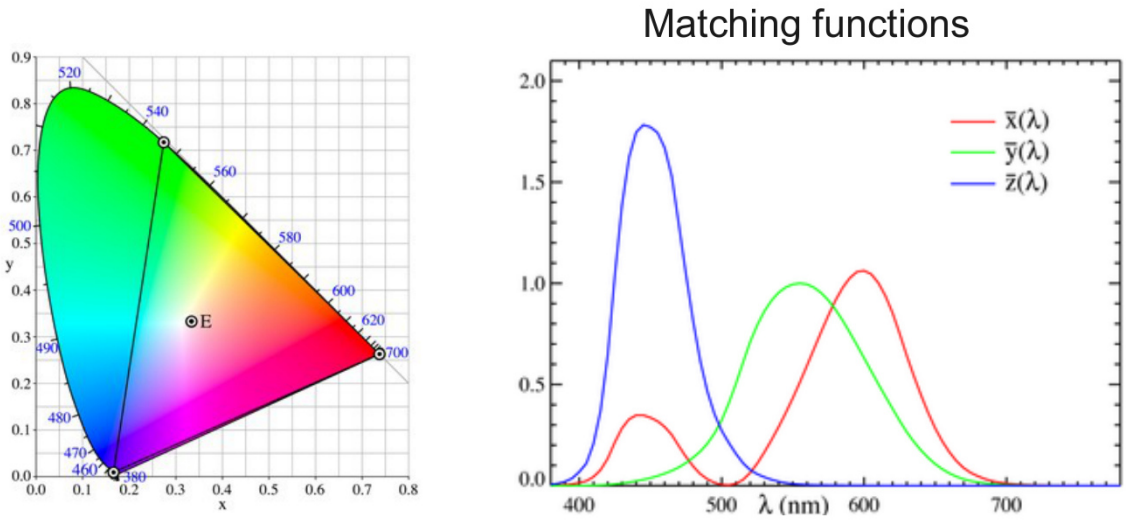
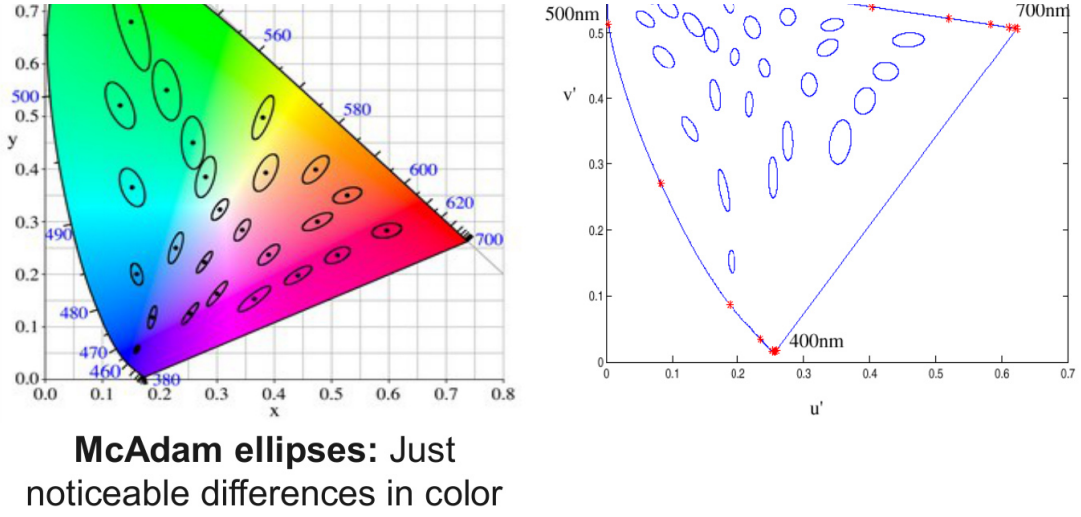


Figure 12.2. CIE XYZ material from the lecture: chromaticity visualization and the associated nonnegative matching functions.



McAdam ellipses: Just noticeable differences in color

Figure 12.3. MacAdam-ellipse illustration used to motivate more perceptually uniform color coordinates.

12.6 HSV

HSV reorganizes color into perceptually interpretable dimensions: **Hue** answers "what color?", **Saturation** describes how much chromatic purity is present, and **Value** describes brightness. The lecture visualizes the model as a cylinder or cone and interprets it geometrically as a reorganization of the RGB cube. In the source's geometric construction, the RGB cube is tilted onto its black–white diagonal: that diagonal becomes the brightness axis, while chromatic colors lie around it. Hue is the angular direction around this axis, saturation moves outward from the neutral axis, and value moves along the brightness direction. The source then schematically places hues in a common plane, expands the cube, and smooths it toward a cylinder. This representation makes operations such as shifting hue or increasing saturation conceptually simple.

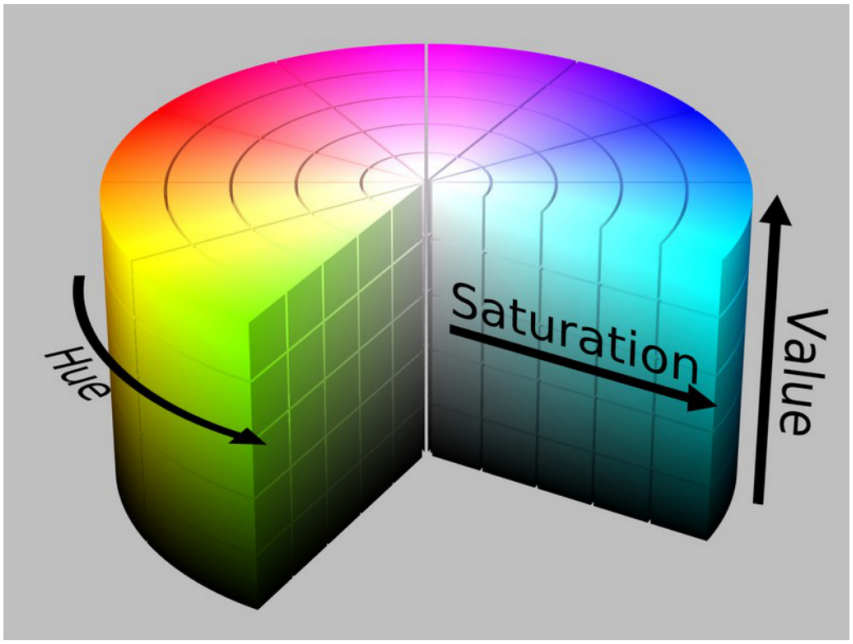


Figure 12.4. Geometric HSV representation shown in the lecture, emphasizing hue, saturation, and value as different axes of description.

Chapter summary. Trichromacy makes three-coordinate color spaces possible. Grassmann's laws justify linear color matching, while $\mathbf{e} = C\mathbf{t}$ maps a sampled spectrum into primary coordinates. RGB is device-oriented, CIE XYZ provides a standardized linear basis, CIE $u'v'$ and $L^*a^*b^*$ address perceptual uniformity, and HSV reorganizes RGB-like color into perceptual dimensions.

CHAPTER 13

Color Constancy and White Balance

13.1 Constancy and Context

Color/lightness constancy is the ability of the human visual system to perceive relatively stable surface reflectance despite changes in illumination. The lecture separates several classes of perceptual effects:

- **instantaneous**: simultaneous contrast and Mach bands;
- **gradual**: light/dark adaptation, chromatic adaptation, and afterimages;
- **high-level**: effects related to color naming and scene interpretation.

Simultaneous contrast means that a target's apparent lightness or color changes with its surrounding field. **Mach bands** emphasize apparent bright and dark bands near transitions between neighboring intensity regions: perceived edge contrast can exceed the physical change.

The checker-shadow illusion shows that equal measured image intensities can be perceived differently. A boundary between checker squares suggests a **reflectance change** (different surface material), whereas the cast-shadow boundary suggests an **illumination change** across a surface. Interpreting the shadow allows the observer to judge surface lightness rather than treating every intensity change as a change of reflectance.

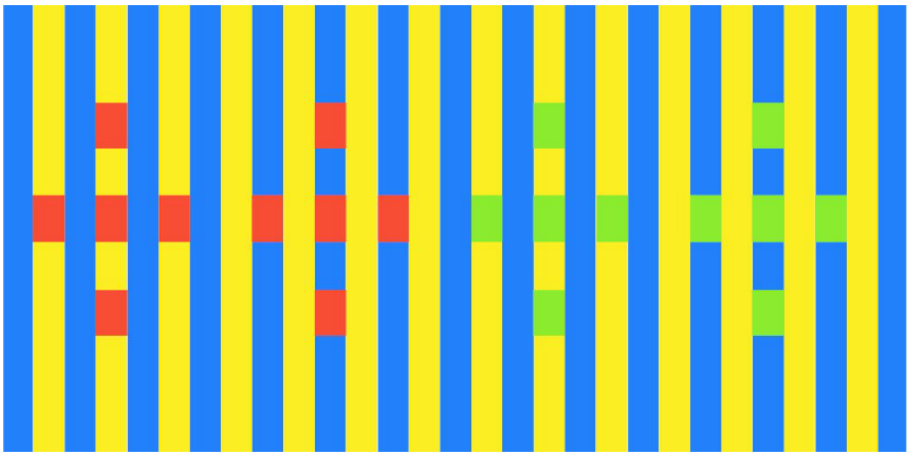


Figure 13.1. The source's simultaneous-contrast/Mach-band demonstration. The colored targets are viewed within alternating surrounding stripes; the perceptual artwork is preserved.

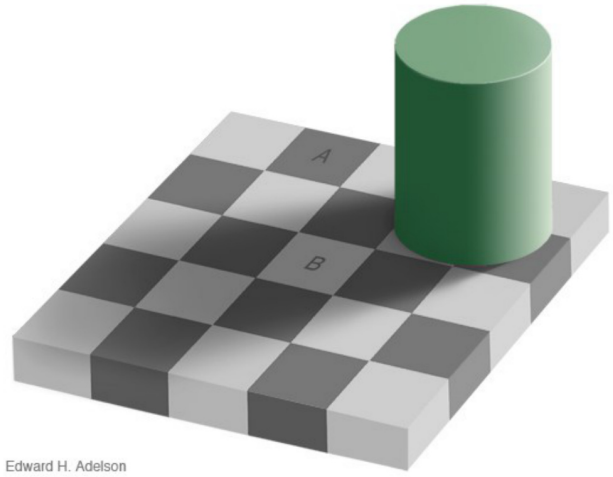


Figure 13.2. Checker-shadow example from the lecture, used to discuss lightness constancy and the distinction between reflectance and illumination changes.

13.2 Chromatic Adaptation

The visual system changes sensitivity according to the prevailing luminance and chromatic conditions. Aperture changes regulate total light entering the eye, while receptor sensitivity also adapts. The lecture's example is that under increased red illumination, red-sensitive responses are gradually reduced until the scene appears more neutral. Adaptation works better in brighter scenes, which helps explain why candlelit scenes may still look yellow.

> **Human perception versus a computer**
>
> Humans use context across the entire scene and perform strong perceptual normalization. A computer ordinarily starts from measured pixel values and does not automatically separate an object's intrinsic color from illumination. Algorithms such as white balance attempt to approximate part of this adaptation.

13.3 White Balance

When a photograph is viewed on a screen or print, the observer adapts to the room's illuminant rather than the illuminant present in the photographed scene. Incorrect camera white balance therefore appears as an unnatural color cast.



Figure 13.3. Incorrect and corrected white balance as illustrated in the lecture.

The lecture contrasts film and digital practice. Film cameras use different films or filters for different illuminants, whereas digital cameras can use automatic white balance, predefined illuminant settings, or a custom neutral reference. Multiple illuminants make the task ambiguous because different reference regions can imply different corrections.

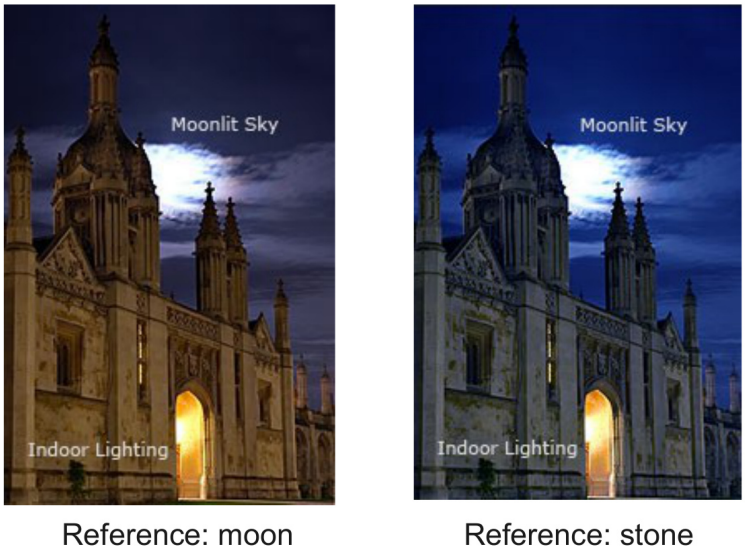


Figure 13.4. Mixed illumination: selecting different neutral references can lead to different white-balance results.

13.3.1 Von Kries Adaptation and a Gray Card

The lecture presents a channel-gain model: multiply each channel by a separate factor. If a known neutral reference is recorded as (r_w, g_w, b_w), choose gains proportional to

$$g_R = \frac{1}{r_w}, \qquad g_G = \frac{1}{g_w}, \qquad g_B = \frac{1}{b_w}. \tag{13.1}$$

where

- (r_w, g_w, b_w): measured RGB values of a white or gray reference object;
- g_R, g_G, g_B: per-channel multiplicative gains (the reference channels must be nonzero).

The slides also note that a light source can have a more complex effect than independent channel scaling; in general it may be represented by a full 3×3 transformation.

13.3.2 White Balance without a Gray Card

When no neutral reference is provided, the camera must infer the illuminant. The lecture lists several strategies:

- **gray-world assumption**: assume the image averages $(r_{ave}, g_{ave}, b_{ave})$ should be gray and use inverse-average gains;
- **brightest-pixel assumption**: highlights often approximate the illuminant color, so use gains inversely proportional to the brightest channel values;
- **gamut mapping**: match the convex hull of observed image colors to a typical gamut under white light;
- **image statistics and learning**: estimate illuminant from learned regularities.

A higher-level method cited in the lecture performs **white balance by recognition**: for semantic classes detected in the image, estimate the illuminant that maps the class's measured average color toward the average color of that class in a database of typical images (Van de Weijer, Schmid, and Verbeek, ICCV 2007).

> **Chapter summary.** Color constancy is a perceptual normalization problem; white balance is its computational analogue in imaging. Simple methods use channel gains and neutral references, while gray-world, brightest-pixel, gamut, statistical, and semantic-class methods estimate the illuminant when no reference card is available.

CHAPTER 14

Color in Computer Vision and Texture

14.1 Uses of Color

The lecture surveys several uses of color in computer vision and associates them with representative research examples:

14.2 Structural and Statistical Texture

The lecture defines **texture** as a description of the spatial arrangement of color or intensities in an image or selected region. A structural approach models a repeated arrangement of basic texture elements (texels). Natural images often make explicit texel extraction difficult or impossible, motivating statistical texture descriptions computed directly from gray tones or colors.

> **Why statistical texture?**
>
> Statistical texture descriptors are less intuitive than explicit texel models but are computationally efficient and can support both classification and segmentation. They are especially useful when natural texture elements cannot be segmented reliably.

Table 14.1. Color-based computer-vision applications and references cited in the lecture.

Application	Representative source named in the lecture
Color histograms for indexing and retrieval	Swain and Ballard, *Color Indexing*, IJCV 1991
Skin detection	Jones and Rehg, *Statistical Color Models with Application to Skin Detection*, IJCV 2002
Image segmentation and retrieval	Carson, Belongie, Greenspan, and Malik, *Blobworld*, 1999
Robot soccer	Sridharan and Stone, color calibration at RoboCup, 2006
Appearance models for tracking	Ramanan, Forsyth, and Zisserman, *Tracking People by Learning their Appearance*, PAMI 2007
Judging visual realism	Lalonde and Efros, color compatibility for image realism, ICCV 2007

14.3 Local Binary Patterns

For each pixel p, the Local Binary Pattern (LBP) example in the lecture compares the eight neighbors with the center value. For neighbor i,

$$b_i = \begin{cases} 0, & \text{if neighbor } i \text{ has value} \le p, \\ 1, & \text{otherwise,} \end{cases} \tag{14.1}$$

Here p denotes the center intensity in this comparison, $i = 1, \ldots, 8$ follows the source neighbor ordering, and b_i is the resulting comparison bit. Then the eight bits $b_1 \ldots b_8$ form a local code. A region is represented by a histogram of these codes.

LBP lecture example

For the 3×3 neighborhood

$$\begin{bmatrix} 100 & 101 & 103 \\ 40 & 50 & 80 \\ 50 & 60 & 90 \end{bmatrix},$$

the center pixel is 50. Starting at the upper-left neighbor and proceeding clockwise gives $100, 101, 103, 80, 90, 60, 50, 40$. Comparing the eight neighbors with the center produces the bit pattern shown in the lecture, 11111100, under its stated neighbor ordering. The histogram of such codes becomes a texture descriptor.

The slides show LBP-histogram retrieval in the Fids image database and explicitly warn that low-level similarity does not always imply semantic similarity. Additional examples mentioned include deciduous-tree classification, finding cancer in Pap smears, and identifying regions of interest in breast and melanoma biopsy slides.

14.4 Image Data Structure

The lecture closes with a simple implementation-oriented image representation:

```
typedef struct{
    int h, w, c;
    float *data;
} image;
```

Here h is height, w is width, and c is the number of channels ($c = 3$ for RGB and $c = 1$ for grayscale in the example). The image data are stored as a linear array of floating-point values in $[0, 1]$, i.e., a three-dimensional image tensor is flattened into one-dimensional memory.

> **Chapter summary.** Color supports retrieval, segmentation, detection, tracking, calibration, and realism assessment. Texture describes spatial appearance; statistical descriptors such as LBP summarize local comparisons as histograms when explicit texel extraction is impractical.

Lecture 3 Quick Review

Light and human vision

1. Color is a perceptual result of physical light interacting with the visual system.
2. Source spectra and surface reflectance jointly determine the spectral signal reaching the observer.
3. Rods are emphasized for intensity sensitivity; three cone classes support color vision.
4. Metamers have different spectra but the same perceived color under the relevant conditions.
5. Trichromacy allows color to be encoded by three primary weights.

Color matching and spaces

6. Grassmann's laws make additive color matching linear.
7. Color matching can be written compactly as $\mathbf{e} = C\mathbf{t}$.
8. RGB is a linear primary space; some spectral matches require subtractive components.
9. CIE XYZ uses nonnegative matching functions and chromaticity coordinates x, y.
10. MacAdam ellipses show that equal x, y distances are not equal perceptual differences.
11. HSV separates hue, saturation, and value for perceptually meaningful manipulation.

Constancy and appearance

12. Color/lightness constancy depends on context and adaptation.
13. White balance compensates for illuminant color; a gray card provides a direct neutral reference.
14. Gray-world, brightest-pixel, gamut, and learning methods estimate white balance without a reference card.
15. Texture describes spatial organization; statistical descriptors avoid explicit texel segmentation.
16. LBP compares neighbors with a center pixel and summarizes the resulting codes with a histogram.

LECTURE 4

Histograms and Intensity Transformations

From intensity distributions to contrast enhancement and normalization.

Histograms · Equalization · Projection · Contrast · Applications

Computer Vision — Marco Raoul Marini

Notes prepared by Matt Clifford Trinidad Ramos

CHAPTER 15

Histograms and Point Transformations

A histogram summarizes how image values are distributed. In this lecture the histogram is used not only as a diagnostic representation of brightness and contrast, but also as the basis for a family of point transformations that remap each input gray level to a new output level.

15.1 Intensity Histograms

For an intensity image, the histogram records how many pixels take each available gray level. The lecture uses the usual display convention in which 0 denotes black and 255 denotes white for an 8-bit image.

Histogram lecture example

The source example uses the 4×5 image

$$\begin{bmatrix} 0 & 1 & 1 & 2 & 4 \\ 2 & 1 & 0 & 0 & 2 \\ 5 & 2 & 0 & 0 & 4 \\ 1 & 1 & 2 & 4 & 1 \end{bmatrix}.$$

Its histogram counts how many times each gray level occurs; in particular the visible bars correspond to counts $5, 6, 5, 0, 3, 1$ for levels $0, 1, 2, 3, 4, 5$.

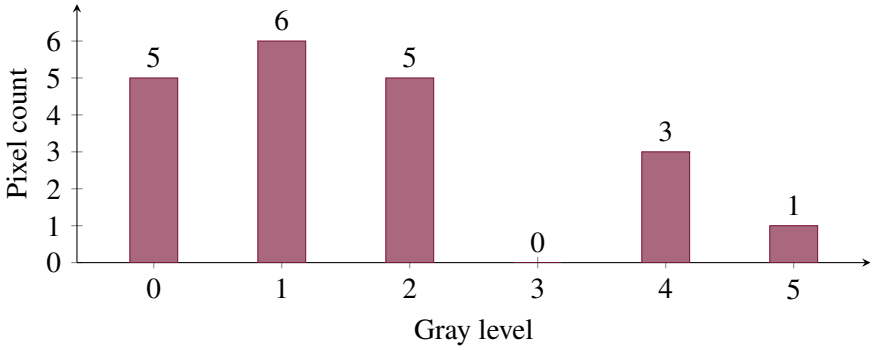


Figure 15.1. Histogram reconstructed from the displayed 4×5 image. The counts sum to 20 pixels.

A histogram is also a direct indicator of contrast. When only a small portion of the available range is occupied, the image has low contrast. The source illustrates this with a gray image whose histogram is concentrated in a narrow interval.

15.2 Histograms of Color Images

The lecture presents three complementary ways to form histograms from color data. First, RGB can be converted to a gray-scale brightness value using

$$Y = 0.299R + 0.587G + 0.114B. \tag{15.1}$$

where

- R, G, B: red, green, and blue channel values;
- Y: gray-scale component in the YIQ color space used in NTSC television;
- the coefficients: weights reflecting the eye's relative brightness sensitivity to the three primaries, as stated in the lecture.

Second, one may compute a separate one-dimensional histogram for each of the red, green, and blue channels. Third, the source notes that a three-dimensional histogram can be formed in RGB space: the three axes represent red, green, and blue, while the value at a location represents the number of pixels with that color combination.

15.3 Point Transformations

Histogram-based enhancement is implemented as a point operation

$$s_k = T(r_k), \tag{15.2}$$

where

- r_k: the k-th input gray level;
- s_k: transformed output gray level;
- T: point transformation applied to gray levels.

The source requires two properties of T:

- it keeps values within the original gray-level range;
- it is monotonically nondecreasing, so the ordering of gray levels is not reversed; distinct input levels may map to the same output level.

> **Why the histogram matters**
>
> A point transformation does not inspect spatial neighborhoods. It changes a pixel only from its current gray level. The histogram therefore supplies exactly the global information needed to design or evaluate many such mappings.

> **Chapter summary.** A histogram counts the occurrence of image values and reveals whether the available intensity range is being used effectively. Color images can be summarized through luminance, separate RGB histograms, or a multidimensional color histogram. Histogram enhancement then applies a monotone point mapping $s_k = T(r_k)$ to redistribute those values.

CHAPTER 16

Histogram Equalization, Projection, and Plateau Methods

16.1 Histogram Equalization

Histogram equalization (HE) transforms image intensities so that the output histogram approximately approaches a flat or uniform distribution. In the discrete formulation shown in the lecture,

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k}\frac{n_j}{n}, \qquad k = 0, 1, \ldots, L-1. \tag{16.1}$$

where

- r_k: the k-th input gray level;
- s_k: equalized output gray level;
- L: number of gray levels in the lecture notation;
- n_j: number of pixels having the j-th gray level;
- n: total number of image pixels;
- $\sum_{j=0}^{k} n_j/n$: cumulative fraction of pixels at or below level k.

> **Source notation for L**
>
> Here L is the number of gray levels: an 8-bit image has $L = 256$ levels, indexed from 0 to $L-1 = 255$. The slide's example "255" denotes the maximum index, not the number of levels. Round the transformed values to valid integer output levels in a discrete implementation.

The cumulative histogram is central: as k increases, the cumulative fraction grows from 0 toward 1, and the transformation spreads frequently occupied input ranges across more of the output interval. Because the image is discrete, the final histogram is only approximately uniform.

The lecture also emphasizes **local equalization**: the histogram can be computed over only part of an image rather than globally. This allows enhancement to adapt to local contrast conditions.

16.2 Histogram Projection

Histogram projection (HP) assigns equal display space to every *occupied* raw signal level, regardless of how many pixels use that level. It therefore depends on which levels are present rather than on their frequencies.

A gray level is occupied if at least one image pixel has that value. Let $B(k)$ denote the fraction of occupied levels at or below level k. If the number of occupied levels is N_{occ}, then $B(k)$ rises from 0 to 1 in uniform steps of $1/N_{\text{occ}}$. The lecture's display transform is

$$s_k = 255\, B(k). \tag{16.2}$$

where

- $B(k)$: cumulative fraction of occupied gray levels at or below k;
- N_{occ}: total number of occupied levels; the slide denotes this quantity by n;
- s_k: display value assigned to raw level k on the 0–255 scale.

> **HE and HP use different counts**
>
> In the HE formula, n denotes the total number of pixels. In the HP slide, n denotes the number of occupied gray levels. The two transformations therefore weight the histogram in fundamentally different ways.

16.3 Plateau Equalization

Plateau equalization (PE) clips histogram counts at a chosen plateau (or saturation) value before forming the cumulative distribution. The resulting display allocation lies between HE and HP.

The source identifies the limiting cases explicitly:

- plateau value = 1: the clipped histogram records only whether a level is occupied, producing $B(k)$ and therefore HP;
- plateau value above the histogram peak: clipping has no effect, producing the ordinary cumulative distribution $F(k)$ and therefore HE;
- intermediate plateau values: produce an intermediate cumulative distribution denoted $P(k)$.

The PE transform is

$$s_k = 255\,P(k), \tag{16.3}$$

where

- $P(k)$: cumulative distribution obtained after clipping histogram counts at the selected plateau value;
- s_k: resulting output display value on the 0–255 scale.

> **Chapter summary.** HE distributes display values according to cumulative pixel frequency, HP distributes them according to occupied gray levels, and PE interpolates between those behaviors by clipping histogram counts before forming the cumulative distribution. Local equalization applies the same idea to a selected image region.

CHAPTER 17

Histogram Specification and Contrast Stretching

17.1 Histogram Specification

Histogram specification (HS), also called histogram matching in the lecture's normalization discussion, transforms an image so that its output histogram approximately matches a desired histogram rather than a uniform one.

Let T denote the cumulative transform of the source histogram and let S denote the cumulative transform associated with the desired histogram. The source diagram expresses the mapping as

$$z = S^{-1}(T(r)), \tag{17.1}$$

where

- r: original image intensity;
- T: cumulative transform of the source histogram;
- S: cumulative transform associated with the desired histogram;
- z: output intensity after applying the inverse target transform.



Figure 17.1. Histogram specification: transform through the source cumulative mapping and the inverse target mapping, corresponding to the lecture's $S^{-1}T$ diagram.

17.2 Contrast Stretching

Contrast stretching (CS) attempts to use the available full gray-level range by expanding a selected input interval. The source gives

$$s_k = 255\,\frac{r_k - r_{\min}}{r_{\max} - r_{\min}}. \tag{17.2}$$

where

- r_k: input gray level;
- $r_{\min}, r_{\max}$: lower and upper values selected for the stretch;
- s_k: output display value on the 0–255 range.

This formula requires $r_{\max} > r_{\min}$. Clamp values outside the selected interval to 0 or 255; a constant image needs an explicit output convention because its minimum and maximum coincide. The same idea can map one intensity interval to another output interval.

The examples show that a simple minimum–maximum stretch is not always enough: when the histogram shape itself is strongly nonuniform, HE can reveal information that CS does not. Conversely, a carefully chosen stretch can preserve a more natural appearance than full equalization.

The source presents several ways to choose the effective input interval:

- use nearly the full distribution, e.g. the 1%–99% percentile range;
- choose explicit limits, illustrated by the pair 79, 136;
- use a **cutoff fraction**, with the slide example 0.8: the cutoff fraction is the minimum fraction of the histogram peak magnitude below which data are ignored when selecting the stretch range.

> **CS versus HE**
>
> Contrast stretching expands a chosen input interval by a direct point mapping. Histogram equalization uses the cumulative histogram and therefore changes the allocation according to how pixels are distributed. The two methods can produce noticeably different results on the same image.

> **Chapter summary.** Histogram specification matches a chosen target distribution using the source and target cumulative transforms. Contrast stretching instead maps a selected input interval across a wider output range. Percentile, explicit-limit, and cutoff-based strategies reduce the influence of extreme or uninformative histogram regions.

CHAPTER 18

Applications of Histogram Processing

18.1 Thresholding

Histogram inspection can support conversion from gray-scale to binary imagery. In a clearly bimodal histogram, a threshold between the two modes can separate the image into two classes. The source's bimodal example uses a threshold of 120.

When the histogram is not bimodal, the choice is less direct. The lecture compares thresholds 80 and 120 on the same example to show that different choices can produce substantially different binary segmentations.

18.2 Normalization and Histogram Matching

When images are to be compared using properties such as texture, the lecture recommends normalizing their histograms to a standard distribution, especially when the images were acquired under different conditions. Histogram equalization is one simple normalization. Histogram matching goes further by taking into account the shapes of both the original histogram and the histogram being matched.

18.3 Medical-Image Examples

The lecture uses CT and MRI to illustrate why histogram processing is useful in practice. In CT lung studies, equalization can be computed over a selected region rather than the full image, emphasizing local diagnostic detail.

MRI intensities do not have a fixed physical meaning in the way the displayed gray value might suggest.

The source states that this is true even for the same protocol, body region, scanner, and patient. Intensity standardization therefore attempts to make images more comparable.

The lecture reproduces the Nyúl–Udupa example: panel A shows histograms of ten FSE proton-density brain-volume images of multiple-sclerosis patients, panel B shows those histograms after scaling, and panel C shows them after final standardization.

18.4 High-Dynamic-Range Display

Digitized high-dynamic-range data such as infrared images and CT/CAT images may span a much larger raw range than the 0–255 levels available on a conventional monitor. A display algorithm must therefore map the raw signal into 0–255 while preserving information that is useful to the human observer.

The source notes two operational examples:

- HP is widely used by infrared camera manufacturers for real-time automated display;
- PE is used in the B-52 infrared navigation and targeting sensor.

> **Chapter summary.** Histogram methods support thresholding, normalization, medical-image standardization, local contrast enhancement, and display of high-dynamic-range data. Their usefulness comes from adapting the available display range to the distribution that matters for the task.

Lecture 4 Quick Review

Histogram foundations

1. A histogram counts how many pixels take each gray level and exposes unused intensity range.
2. RGB may be converted to brightness by $Y = 0.299R + 0.587G + 0.114B$, or analyzed with separate/multidimensional color histograms.
3. Histogram point processing uses a monotone mapping $s_k = T(r_k)$ within the available range.

Equalization family

4. HE maps cumulative pixel frequency across the display range and can also be applied locally.
5. HP assigns equal display space to every occupied gray level, independent of its pixel count.
6. PE clips histogram peaks: plateau 1 gives HP; a plateau above the peak gives HE.

Specification and contrast

7. Histogram specification uses $S^{-1}(T(r))$ to approximately match a target distribution.
8. Contrast stretching maps a selected interval to a wider range; the source also uses percentile, fixed-limit, and cutoff-fraction choices.
9. CS and HE need not produce the same enhancement because only HE uses cumulative pixel frequency.

Applications

10. Histogram structure can guide threshold selection; non-bimodal cases are more ambiguous.
11. Normalization makes differently acquired images more comparable; histogram matching accounts for both histogram shapes.
12. MRI intensity standardization, CT local enhancement, and IR/CT display are representative applications.

LECTURE 5

Spatial Filters and Image Enhancement

From local neighborhoods to smoothing, sharpening, and combined enhancement.

Neighborhoods · Smoothing · Laplacian · Sobel · Enhancement

Computer Vision — Marco Raoul Marini

Notes prepared by Matt Clifford Trinidad Ramos

CHAPTER 19

Spatial Filtering Foundations

Lecture 5 moves from point operations to **neighborhood operations**. Instead of transforming a pixel from its value alone, a spatial filter computes the output at (x, y) from a neighborhood around that location. The source follows the spatial-filtering material of Gonzalez and Woods, *Digital Image Processing*, Chapter 3.

19.1 Neighborhood Operations

A neighborhood is typically a rectangular region centered on the current pixel, although the lecture notes that other sizes and shapes are possible. For each pixel of the original image, an output is written at the corresponding location in the target image.

Two simple non-linear neighborhood examples are:

- **minimum filter**: replace the center pixel by the minimum value in its neighborhood;
- **maximum filter**: replace the center pixel by the maximum value in its neighborhood.

19.2 The Spatial Filtering Process

For a filter of size $(2a + 1) \times (2b + 1)$, the lecture gives the correlation-style spatial filtering equation

$$g(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x + s, y + t). \tag{19.1}$$

where

- $f(x, y)$: input image;
- $g(x, y)$: filtered output image;
- $w(s, t)$: filter/kernel coefficient at offset (s, t);
- a, b: half-sizes of the neighborhood in the two directions;
- s, t: offsets relative to the current pixel.

The operation is repeated at every pixel location. The source's 3×3 diagram writes the processed center as the sum of the nine neighboring image samples multiplied by the corresponding nine kernel entries.

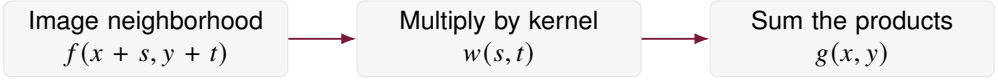


Figure 19.1. Correlation at one pixel: multiply corresponding samples and kernel weights, then sum. Repeat at every output location.

19.3 Correlation and Convolution

The filtering operation used up to this point is called **correlation**, and the filter is a correlation kernel. **Convolution** is the closely related operation obtained by reversing the kernel before applying the same local multiply-and-sum process. The lecture emphasizes that the distinction disappears for symmetric filters.

> **Correlation versus convolution**
>
> If a kernel is symmetric under the required reversal, correlation and convolution produce the same result. This is why many smoothing kernels behave identically under the two operations even though the formal definitions differ.

19.4 Image Borders

At an image boundary, a full neighborhood may extend beyond the available pixels. The source lists four strategies:

- omit missing pixels when the filter permits it;
- pad the image, typically with all-white or all-black pixels;
- replicate border pixels;
- truncate the output image.

> **Chapter summary.** Spatial filtering computes each output pixel from a local neighborhood. Correlation applies the kernel as written; convolution reverses it first, with no difference for symmetric kernels. Boundary handling must be specified because complete neighborhoods are unavailable at image edges.

CHAPTER 20

Smoothing Spatial Filters

20.1 Averaging Filters

The simplest smoothing filter replaces the center pixel by the average of the pixels in its neighborhood. For a 3×3 neighborhood, the source uses

$$W_{\text{mean}} = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Smoothing is useful for reducing noise and for suppressing fine detail so that large-scale structures become more prominent.

Numerical averaging example from the lecture

For the neighborhood

$$\begin{bmatrix} 104 & 100 & 108 \\ 99 & 106 & 98 \\ 95 & 90 & 85 \end{bmatrix},$$

the mean-filter output at the center is

$$\frac{104 + 100 + 108 + 99 + 106 + 98 + 95 + 90 + 85}{9} = 98.3333.$$

The same computation is repeated over the image.

As the averaging-filter size increases, more fine detail disappears. The source compares filter sizes 3, 5, 9, 15, and 35 on a 500×500 test image.

20.2 Weighted Smoothing

More selective smoothing assigns different importance to different neighbors. The lecture's weighted averaging example is

$$W_{\text{weighted}} = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

Pixels closer to the center receive greater weight. The coefficients sum to one, so a constant region remains unchanged by the filter.

20.3 Smoothing as Preprocessing

By removing fine detail, smoothing can simplify a later decision such as thresholding. The source illustrates this with an astronomical image: smoothing suppresses small bright structures and leaves the gross objects that are then isolated by thresholding.

20.4 Averaging and Median Filtering

Filtering is frequently used to remove noise, but the lecture stresses that an averaging filter is not always the best choice. Its comparison shows that a **median filter** can preserve structures while removing impulsive-looking noise more effectively than a simple average.

Chapter summary. Averaging filters reduce noise and fine structure by local averaging, while weighted smoothing gives greater influence to selected neighbors. Increasing the support size produces stronger blur. Median filtering is a non-linear alternative that can outperform averaging on some noise patterns.

CHAPTER 21

Sharpening and Spatial Derivatives

Sharpening filters seek to emphasize fine detail, reduce the visual effect of blur, and highlight edges. The lecture develops them from spatial differentiation.

21.1 First and Second Discrete Derivatives

For a one-dimensional function, the source uses the forward first difference

$$\frac{\partial f}{\partial x} = f(x+1) - f(x), \tag{21.1}$$

and the second difference

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x). \tag{21.2}$$

where

- $f(x)$: one-dimensional sampled signal;
- x: sample position;
- $\partial f/\partial x$: first discrete difference used in the lecture;
- $\partial^2 f/\partial x^2$: second discrete difference.

The first derivative measures the change between successive values, while the second derivative compares the current value with both its predecessor and successor.

The source summarizes the practical differences as follows:

- first-order derivatives generally produce thicker edges;
- second-order derivatives respond more strongly to fine detail such as thin lines;
- first-order derivatives have a stronger response to a gray-level step;
- second-order derivatives produce a double response at a gray-level step.

The second derivative is emphasized for image enhancement because of its strong response to fine detail and simple implementation.

21.2 The Laplacian

The Laplacian is the isotropic second-order operator

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}. \tag{21.3}$$

With the source's discrete second differences,

$$\begin{aligned} \nabla^2 f(x,y) = {} & f(x+1,y) + f(x-1,y) \\ & + f(x,y+1) + f(x,y-1) - 4f(x,y). \end{aligned} \tag{21.4}$$

where

- $f(x, y)$: image intensity at spatial coordinates (x, y);
- $\partial^2 f/\partial x^2$ and $\partial^2 f/\partial y^2$: second differences in the horizontal and vertical directions;
- $\nabla^2 f$: Laplacian response obtained by summing those second-order components.

This corresponds to the kernel

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Applying the Laplacian alone produces an image dominated by edges and other discontinuities; it is therefore not yet the final enhanced image. With the sign convention above, the lecture sharpens by subtracting the Laplacian response:

$$g(x, y) = f(x, y) - \nabla^2 f(x, y). \tag{21.5}$$

where

- $f(x, y)$: original image;
- $\nabla^2 f(x, y)$: Laplacian response;
- $g(x, y)$: sharpened image.

The subtraction can be combined into one spatial filter:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}.$$

The source also shows the 8-neighbor Laplacian variant

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

and its corresponding one-step sharpening kernel

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix}.$$

21.3 Unsharp Masking and Highboost Filtering

The lecture presents unsharp masking as a sequence of linear spatial operations:

1. blur the original image;
2. subtract the blurred image from the original to obtain an unsharp mask;
3. add the mask back to the original to obtain the sharpened image.

The following source illustration also compares the result with a stronger highboost-filtered image.

21.4 First-Derivative Filtering and the Gradient

For a two-dimensional image, the gradient is

$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \partial f/\partial x \\ \partial f/\partial y \end{bmatrix}. \tag{21.6}$$

Its magnitude is

$$\|\nabla f\| = \sqrt{G_x^2 + G_y^2}, \tag{21.7}$$

and the lecture gives the practical approximation

$$\|\nabla f\| \approx |G_x| + |G_y|. \tag{21.8}$$

where

- $G_x = \partial f/\partial x$: horizontal gradient component;
- $G_y = \partial f/\partial y$: vertical gradient component;
- ∇f: gradient vector;
- $\|\nabla f\|$: gradient magnitude; the lecture also uses the L_1-style approximation $|G_x| + |G_y|$.

Using the 3×3 neighborhood

$$\begin{bmatrix} z_1 & z_2 & z_3 \\ z_4 & z_5 & z_6 \\ z_7 & z_8 & z_9 \end{bmatrix},$$

the source writes the magnitude approximation as

$$\begin{aligned} \|\nabla f\| \approx\ & |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| \\ & + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|\,. \end{aligned} \tag{21.9}$$

where

- $z_1, \ldots, z_9$: the nine image samples in the displayed 3×3 neighborhood;
- the first absolute difference: one Sobel directional response;
- the second absolute difference: the orthogonal Sobel directional response.

21.5 Sobel Operators

The preceding expression leads to the two Sobel masks shown in the lecture:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \qquad \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}.$$

The image is filtered with both operators and the two responses are combined. Sobel filtering is typically used for edge detection.

> **Chapter summary.** Sharpening is based on spatial derivatives. The Laplacian uses second differences and can be subtracted from the image for sharpening; unsharp masking forms a high-frequency mask by subtracting a blur; and first-derivative methods use the image gradient, with Sobel

masks providing a practical approximation to horizontal and vertical changes.

CHAPTER 22

Combining Spatial Enhancement Methods

The final lecture example stresses that successful enhancement is typically not achieved by a single operator. Several filters and point transformations can be combined so that one stage supplies information needed by the next.

22.1 Bone-Scan Enhancement Pipeline

The source combines Laplacian sharpening, Sobel edges, smoothing, masking, and a final power-law transformation on a bone scan. The illustrated stages are:

1. **(a)** original bone scan;
2. **(b)** Laplacian-filtered version of (a);
3. **(c)** sharpened image obtained by subtracting the Laplacian result from the original;
4. **(d)** Sobel-filtered version of (a);
5. **(e)** image (d) smoothed with a 5×5 averaging filter;
6. **(f)** product of the sharpened image (c) and the smoothed Sobel image (e), used as a mask;
7. **(g)** sharpened image formed as the sum of the original (a) and mask (f);
8. **(h)** result of applying a power-law transformation to (g).

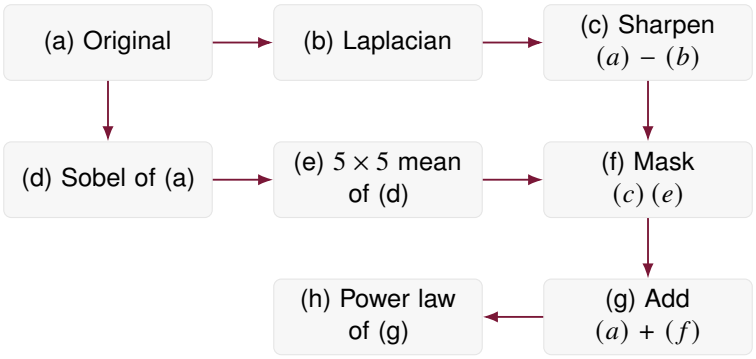


Figure 22.1. Dependencies of the bone-scan enhancement stages described above. The original (a) also supplies the subtraction in (c) and addition in (g).

Enhancement as a pipeline

The example demonstrates the purpose of combining operators: the Laplacian supplies fine-detail sharpening, the Sobel operator supplies edge information, smoothing stabilizes that edge map, multiplication forms a spatially selective mask, and the final point transformation adjusts the visual result.

Chapter summary. Spatial enhancement can combine complementary operations rather than relying on one filter. The bone-scan example uses Laplacian sharpening, Sobel edge detection, averaging, multiplicative masking, addition to the original, and a final power-law transformation to obtain the final image.

Lecture 5 Quick Review

Spatial filtering

1. Neighborhood operations use local pixels rather than a single input value.
2. Linear spatial filtering computes a weighted sum $g(x, y) = \sum\sum w(s, t) f(x + s, y + t)$.
3. Correlation uses the kernel as written; convolution reverses it. Symmetric kernels give the same result.
4. Borders require a policy such as omission, padding, replication, or truncation.

Smoothing

5. Mean filtering averages a neighborhood; larger supports remove more fine detail.
6. Weighted smoothing emphasizes selected neighbors, typically giving the center more influence.
7. Smoothing can expose gross structures for later thresholding; median filtering can outperform averaging on some noise.

Sharpening

8. First differences detect change; second differences respond strongly to fine detail and produce a double response at gray-level steps.
9. The Laplacian is $\nabla^2 f = f_{xx} + f_{yy}$; with the source kernel, sharpening uses $g = f - \nabla^2 f$.
10. Unsharp masking subtracts a blur to form a high-frequency mask and adds that mask to the original.
11. The gradient magnitude is $\sqrt{G_x^2 + G_y^2}$ and is often approximated by $|G_x| + |G_y|$.
12. Sobel filtering uses two 3×3 derivative masks and is typically used for edge detection.

Combined enhancement

13. The lecture's bone-scan example combines Laplacian sharpening, Sobel edges, averaging, masking, addition, and a power-law transform.
14. Different operators can be chained because they emphasize complementary image information.